PRIMITIVE RECURSIVE ARITHMETIC

1. Introduction

In these notes we develop primitive recursive arithmetic. Although our development can be fully formalized in the first-order logic, we adopt a style analogous to set theory textbooks: rigorous but expressed using natural-language arguments rather than syntactical intricacies. The main difference is that, unlike set theory, our universe of discourse consists of natural numbers rather than sets. Consequently, all the theories we introduce share the feature that the range of their quantifiers is the set of natural numbers. Following Quine's principle that quantifying carries ontological commitment we are in these notes committed to the existence of natural numbers.

2. First-order logic

We denote object-language variables by

$$n, m, k, ...$$

In these notes variables range over *natural numbers*. We also use

$$\mathrm{n}, \mathrm{m}, \mathrm{k}, ...$$

to denote natural numbers at meta-level. For example we write $\vec{n}$ for a tuple $(n_1, ..., n_{\mathrm{n}})$ of object-language variables of length n, where n is some meta-level natural number.

We operate within classical deductive system introduced in [**?**]. Free variables in formulas are treated as parameters. No implicit universal closure is ever assumed. We implicitly use alphabetic variants and do not bother with the usual issues related to substitution. Moreover, we apply conservativity results described in [**?**]. Specifically, if $\mathcal{L}$ is a first-order language and Γ is a collection of formulas of $\mathcal{L}$, then we freely extend $\mathcal{L}$ and Γ as follows.

1. For every n-tuple $\vec{n}$ of variables and a formula $\Phi(\vec{n})$ we may add a fresh n-ary relation symbol R to $\mathcal{L}$ and extend Γ by adding

$$\forall_{\vec{n}}\ (\Phi(\vec{n}) \Leftrightarrow R(\vec{n}))$$

2. For every n-tuple $\vec{n}$ of variables and a formula $\Phi(\vec{n}, m)$ such that

$$\Gamma \vdash \forall_{\vec{n}} \exists!_m\, \Phi(\vec{n}, m)$$

we may add a fresh n-ary function symbol F to $\mathcal{L}$ and extend Γ by adding

$$\forall_{\vec{n}}\, \Phi(\vec{n}, F(\vec{n}))$$

3. For every n-tuple $\vec{n}$ of variables and a formula $\Phi(\vec{n}, m)$ such that

$$\Gamma \vdash \exists_m\, \Phi(\vec{n}, m)$$

we may add a fresh constant symbol c to $\mathcal{L}$ and extend Γ by adding $\Phi(\vec{n}, c)$.

In these notes a theorem, proposition, fact, lemma or corollary labeled by a collection of formulas Γ denotes a statement derivable in Γ and the accompanying proof is intended to be formalizable in Γ. Unlabeled theorems are meta-theorems.

3. The system of primitive recursive arithmetic

In this section we introduce the system PRA. The language of PRA consists only of function symbols. It contains base symbols

Zero symbols.

$\underbrace{\mathbf{Z}_{\mathrm{k}}}_{\mathrm{k-ary}}$ for every arity k. Also $\mathbf{Z}_0$ is denoted by 0.

The successor.

$\underbrace{\mathbf{S}}_{\mathrm{unary}}$

Projection symbols.

$\underbrace{\mathbf{I}^{\mathrm{i}}_{\mathrm{k}}}_{\mathrm{k-ary}}$ for every nonzero arity k and $1 \leq \mathrm{i} \leq \mathrm{k}$.

and is defined inductively by the following operations.

Composition.

If G is a k-ary symbol and $F_1, ..., F_{\mathrm{k}}$ are m-ary symbols, then $\mathrm{comp}(G, F_1, ..., F_{\mathrm{k}})$ is a m-ary symbol. We say that $\mathrm{comp}(G, F_1, ..., F_{\mathrm{k}})$ is obtained from $G, F_1, ..., F_{\mathrm{k}}$ by *composition*.

Primitive recursion.

If G is $(\mathrm{k}+2)$-ary symbol and F is a k-ary symbol, then $\mathrm{prim}(G, F)$ is a $(\mathrm{k}+1)$-ary symbol. We say that $\mathrm{prim}(G, F)$ is obtained from G, F by *primitive recursion*.

Definition 3.1. Function symbols of the language of PRA are called *primitive recursive* functions.

The nonlogical axioms of PRA are universal closures of the following formulas.

1. $\mathbf{Z}_{\mathrm{k}}(\vec{n}) = 0$ for k-tuple $\vec{n}$ of variables.
2. $\mathbf{S}(n) \neq 0$
3. $\mathbf{I}^{\mathrm{i}}_{\mathrm{k}}(\vec{n}) = n_{\mathrm{i}}$ for k-tuple $\vec{n}$ of variables and all $1 \leq \mathrm{i} \leq \mathrm{k}$.
4. If G is a k-ary symbol and $F_1, ..., F_{\mathrm{k}}$ are m-ary symbols, then
$$\mathrm{comp}(G, F_1, ..., F_{\mathrm{k}})(\vec{n}) = G\left(F_1(\vec{n}), ..., F_{\mathrm{k}}(\vec{n})\right)$$
for m-tuple $\vec{n}$ of variables.
5. If G is $(\mathrm{k}+2)$-ary symbol and F is a k-ary symbol, then
$$\mathrm{prim}(G, F)(\vec{n}, 0) = F(\vec{n})$$
and
$$\mathrm{prim}(G, F)\left(\vec{n}, S(m)\right) = G\left(\vec{n}, m, \mathrm{prim}(G, F)(\vec{n}, m)\right)$$
where $\vec{n}$ is a k-tuple of variables and n is a variable distinct from $\vec{n}$.
6. **Induction axiom schema.**

   Suppose that $\Phi(\vec{n}, m)$ is a quantifier-free formula in the language of PRA. Then
$$\big(\Phi(\vec{n}, 0) \wedge \forall_m \left(\Phi(\vec{n}, m) \Rightarrow \Phi(\vec{n}, S(m))\right)\big) \Rightarrow \forall_m \Phi(\vec{n}, m)$$

In addition to this nonlogical axioms PRA has usual first-order axioms with identity.

From now on unless otherwise stated formulas are always understood to be formulas of the language of PRA.

4. Rudimentary results in primitive recursive arithmetic

The reader my find this section rather dry and tedious. The goal here is to show that the most rudimentary facts on natural numbers can be proved in PRA. First we introduce familiar notation.

Definition 4.1. $1 = \mathbf{S}(0), 2 = \mathbf{S}(\mathbf{S}(0)), ...$

Definition 4.2. $\mathbf{P}(0) = 0$, $\mathbf{P}(\mathbf{S}(n)) = n$.

Corollary 4.3 [PRA]. *The function* $\mathbf{S}$ *is injective.*

Proof. If $\mathbf{S}(n) = \mathbf{S}(m)$, then $n = \mathbf{P}(\mathbf{S}(n)) = \mathbf{P}(\mathbf{S}(m)) = m$. □

Corollary 4.4 [PRA]. *If $n \neq 0$, then there exists m such that $\mathbf{S}(m) = n$.*

Proof. We prove the assertion by induction on n. For $n = 0$ it is clear. Suppose that it holds for some n. Then

$$\mathbf{S}\left(\mathbf{P}\left(\mathbf{S}(n)\right)\right) = \mathbf{S}(n)$$

and hence it holds for $\mathbf{S}(n)$. Thus formula in question is universally valid. □

Next we introduce addition and multiplication.

Definition 4.5. $n + 0 = n$, $n + \mathbf{S}(m) = \mathbf{S}(n + m)$

Definition 4.6. $n \cdot 0 = 0$, $n \cdot \mathbf{S}(m) = n \cdot m + n$

Theorem 4.7 [PRA]. *The following assertions hold.*

- **(1)** $n + m = m + n$
- **(2)** $(n + m) + k = n + (m + k)$
- **(3)** $n + k = m + k \Rightarrow n = m$
- **(4)** $n + m = 0 \Rightarrow (n = 0 \wedge m = 0)$
- **(5)** $n \cdot (m + k) = x \cdot m + n \cdot k$
- **(6)** $n \cdot m = m \cdot n$
- **(7)** $(n \cdot m) \cdot k = n \cdot (m \cdot k)$
- **(8)** $n \cdot m = 0 \Rightarrow (n = 0 \wedge m = 0)$

Proof. We leave the proof as the exercise. □

Now we introduce ordering relation.

Definition 4.8. $n \leq m \equiv \exists_k \, m = k + n$

Definition 4.9. $n < m \equiv n \leq m \wedge n \neq m$

In order to prove that ordering relation is quantifier-free we need truncated subtraction function.

Definition 4.10. $n \dot{-} 0 = n$, $n \dot{-} \mathbf{S}(m) = \mathbf{P}\left(n \dot{-} m\right)$

Proposition 4.11 [PRA]. *The following assertions are equivalent.*

- **(i)** $n \leq m$

(ii) $m = (m \dot{-} n) + n$

For the proof we need the following result.

Lemma 4.11.1 [PRA]. *The following assertions hold.*

(1) $\mathbf{S}(m) \dot{-} \mathbf{S}(n) = m \dot{-} n$

(2) $(k + n) \dot{-} n = k$ *for every* n.

Proof of the lemma. The proof of **(1)** goes by induction on n. For base case we have

$$\mathbf{S}(m) \dot{-} \mathbf{S}(0) = \mathbf{P}(\mathbf{S}(m) \dot{-} 0) = \mathbf{P}(\mathbf{S}(m)) = m = m \dot{-} 0$$

Suppose that the assertion holds for some n. Then

$$\mathbf{S}(m) \dot{-} \mathbf{S}(\mathbf{S}(n)) = \mathbf{P}(\mathbf{S}(m) \dot{-} \mathbf{S}(n)) \underbrace{=}_{\text{induction hypothesis}} \mathbf{P}(m \dot{-} n) = m \dot{-} \mathbf{S}(n)$$

This proves that **(1)** holds.

The proof of **(2)** goes by induction on n. The base case is $k \dot{-} 0 = k$ and this follows by definition of $\dot{-}$. Suppose that $(k + n) \dot{-} n = k$ for some n. Then

$$(k + \mathbf{S}(n)) \dot{-} \mathbf{S}(n) = \mathbf{S}(k + n) \dot{-} \mathbf{S}(n) \underbrace{=}_{\textbf{(1)}} (k + n) \dot{-} n \underbrace{=}_{\text{induction hypothesis}} k$$

This completes the proof. □

Proof of the proposition. Suppose that **(i)** holds. Then there exists k such that $m = k + n$. We have

$$k \underbrace{=}_{\text{Lemma 4.11.1}} (k + n) \dot{-} n = m \dot{-} n$$

and hence $m = (m \dot{-} n) + n$. This is **(ii)**.

The implication **(ii)** $\Rightarrow$ **(i)** is obvious. □

Theorem 4.12 [PRA]. *The following assertions hold.*

(1) $0 \leq n$

(2) $n \leq n$

(3) $n < m \Leftrightarrow \exists_k \, (k \neq 0 \wedge n = k + n)$

(4) $(n \leq m \wedge m \leq k) \Rightarrow n \leq k$

(5) $(n \leq m \vee m \leq x)$

(6) $n \leq m \Rightarrow x \cdot k \leq m \cdot k$

(7) $n \leq m \Rightarrow n + k \leq m + k$

Proof. According to Proposition 4.11 formula $n \leq m$ is equivalent to a quantifier-free formula. We only prove assertion **(4)** and leave the rest for the reader. Fix n as a parameter and prove by induction on m that $n \leq m$ or $m \leq n$. Note that $0 \leq n$ according to **(1)**. This proves the base case. Suppose that the assertion holds for some m. We consider cases.

- If $n \leq m$, then there exists z such that $m = k + n$. Note that

$$\mathbf{S}(m) = \mathbf{S}(k + n) \underbrace{=}_{\text{Theorem 4.7}} \mathbf{S}(n + k) = n + \mathbf{S}(k) \underbrace{=}_{\text{Theorem 4.7}} \mathbf{S}(k) + n$$

and hence $n \leq \mathbf{S}(m)$.

- If $m = n$, then
$$\mathbf{S}(m) = \mathbf{S}(n) = n + 1 \underbrace{=}_{\text{Theorem 4.7}} 1 + n$$
and hence $n \leq \mathbf{S}(m)$.
- If $m \leq n$ and $m \neq n$, then by definition $m < n$. Hence by **(3)** there exists $k \neq 0$ such that $n = k + m$. Corollary 4.4 implies that there exists l such that $k = \mathbf{S}(l)$. Thus
$$n = k + m = \mathbf{S}(l) + m \underbrace{=}_{\text{Theorem 4.7}} m + \mathbf{S}(l) = \mathbf{S}(m + l) \underbrace{=}_{\text{Theorem 4.7}} \mathbf{S}(l + m) = l + \mathbf{S}(m)$$
This proves that $\mathbf{S}(m) \leq n$.

Thus $n \leq \mathbf{S}(m)$ or $\mathbf{S}(m) \leq n$. Hence **(4)** follows by induction on m. □

5. Primitive recursive formulas and characteristic functions

This section is devoted to important meta-theorems.

Although it is very basic, it is worth noting, in order to make presentation rigorous that the composition of $\mathbf{S}$ and the zero function gives rise for every k to the constant k-ary function that admits value 1. By convention we denote this function by 1 and assume that its use is clear from the context.

Definition 5.1. $\text{sign}(0) = 0$, $\text{sign}(\mathbf{S}(n)) = 1$

Definition 5.2. Let $\Phi(\vec{n})$ be a formula and let C be a primitive recursive function. Suppose that
$$\text{PRA} \vdash \forall_{\vec{n}} \left(C(\vec{n}) = 0 \vee C(\vec{n}) = 1 \right) \wedge \forall_{\vec{n}} \left(\Phi(\vec{n}) \Leftrightarrow C(\vec{n}) = 1 \right)$$
Then C is a *characteristic function* of $\Phi(\vec{n})$.

Definition 5.3. A formula $\Phi(\vec{n})$ that admits a characteristic function is *primitive recursive*.

Theorem 5.4. *Let $\Phi(\vec{n})$ be a formula. Then the following assertions are equivalent.*

(i) *$\Phi(\vec{n})$ is primitive recursive.*

(ii) *There exists a quantifier-free formula $\Theta(\vec{n})$ such that*
$$\text{PRA} \vdash \Phi(\vec{n}) \Leftrightarrow \Theta(\vec{n})$$

Proof. The implication **(i)** $\Rightarrow$ **(ii)** is straightforward.

We construct a characteristic function C_Φ for every quantifier-free formula $\Phi(\vec{n})$. The construction goes by induction on the complexity of $\Phi(\vec{n})$.

(1) $\Phi(\vec{n})$ is of the form $F_1(\vec{n}) = F_2(\vec{n})$ where F_1, F_2 are function symbols. Then we define
$$C_\Phi(\vec{n}) = 1 \dot{-} \text{sign}\big(F_1(\vec{n}) \dot{-} F_2(\vec{n}) + F_2(\vec{n}) \dot{-} F_1(\vec{n})\big)$$
Clearly the right hand side is defined in terms of composition from sign, $+, \dot{-}, F_1, F_2$.

We prove that C_Φ is a characteristic function for $\Phi(\vec{n})$. Clearly $C_\Phi(\vec{n}) = 0, 1$ for all $\vec{n}$. Moreover, by definition of $\dot{-}$ and sign the formula $C_\Phi(\vec{n}) = 1$ is equivalent to
$$F_1(\vec{n}) \dot{-} F_2(\vec{n}) + F_2(\vec{n}) \dot{-} F_1(\vec{n}) = 0$$
By Proposition 4.11 and Theorem 4.12 this last formula is provably equivalent in PRA to $F_1(\vec{n}) = F_2(\vec{n})$.

(2) $\Phi(\vec{n})$ is $\Psi_1(\vec{n}) \vee \Psi_2(\vec{n})$ for formulas $\Psi_1(\vec{n})$ and $\Psi_2(\vec{n})$ with characteristic functions $C_{\Psi_1}(\vec{n})$ and $C_{\Psi_2}(\vec{n})$, respectively. Then

$$C_{\Phi}(\vec{n}) = \mathrm{sign}\left(C_{\Psi_1}(\vec{n}) + C_{\Psi_2}(\vec{n})\right)$$

is a characteristic function of $\Phi(\vec{n})$.

(3) $\Phi(\vec{n})$ is $\Psi_1(\vec{n}) \wedge \Psi_2(\vec{n})$ for formulas $\Psi_1(\vec{n})$ and $\Psi_2(\vec{n})$ with characteristic functions $C_{\Psi_1}(\vec{n})$ and $C_{\Psi_2}(\vec{n})$, respectively. Then

$$C_{\Phi}(\vec{n}) = C_{\Psi_1}(\vec{n}) \cdot C_{\Psi_2}(\vec{n})$$

is a characteristic function of $\Phi(\vec{n})$.

(4) $\Phi(\vec{n})$ is $\neg\Psi(\vec{n})$ for some formula $\Psi(\vec{n})$ with characteristic function $C_{\Psi}(\vec{n})$. Then

$$C_{\Phi}(\vec{n}) = 1 \dot{-} C_{\Psi}(\vec{n})$$

is a characteristic function of $\Phi(\vec{n})$.

(5) $\Phi(\vec{n})$ is $\Psi_1(\vec{n}) \Rightarrow \Psi_2(\vec{n})$ for formulas $\Psi_1(\vec{n})$ and $\Psi_2(\vec{n})$ with characteristic functions $C_{\Psi_1}(\vec{n})$ and $C_{\Psi_2}(\vec{n})$, respectively. Since $\Psi_1(\vec{n}) \Rightarrow \Psi_2(\vec{n})$ is logically equivalent to $(\neg\Psi_1(\vec{n})) \vee \Psi_2(\vec{n})$ this case reduces to previously considered cases.

Thus each quantifier-free formula admits a characteristic function. Since the collection of formulas that admit characteristic functions is closed under equivalence in PRA, we infer that the each formula provably equivalent in PRA to quantifier-free formula admits a characteristic function. This completes the proof of **(ii)** $\Rightarrow$ **(i)**. □

Corollary 5.5. *The collection of primitive recursive formulas is closed under* $\wedge, \vee, \Rightarrow, \neq$.

Proof. By Theorem 5.4 primitive recursive formulas are provably equivalent in PRA to quantifier-free formulas. By definition quantifier-free fornulas are closed under $\wedge, \vee, \Rightarrow, \neq$. Hence primitive recursive formulas are closed under $\wedge, \vee, \neg, \Rightarrow$. □

Corollary 5.6. *Let* F_{i} *be a* k*-ary primitive recursive function for* $\mathrm{i} = 1, ..., \mathrm{n}$. *Suppose that* $\vec{n}$ *is a* k*-tuple of variables. Let* $\Phi_{\mathrm{i}}(\vec{n})$ *be primitive recursive formulas for* $\mathrm{i} = 1, ..., \mathrm{n}$ *such that the following assertions hold.*

(1) $\mathrm{PRA} \vdash \forall_{\vec{n}}\left(\Phi_1(\vec{n}) \vee ... \vee \Phi_{\mathrm{n}}(\vec{n})\right)$

(2) $\mathrm{PRA} \vdash \forall_{\vec{n}}\left(\neg\left(\Phi_{\mathrm{i}}(\vec{n}) \wedge \Phi_{\mathrm{j}}(\vec{n})\right)\right)$ *for all* $\mathrm{i} \neq \mathrm{j}$ *such that* $\mathrm{i}, \mathrm{j} = 1, ..., \mathrm{n}$.

Then there exists k*-ary primitive recursive function* F *such that*

$$\mathrm{PRA} \vdash \forall_{\vec{n}}\left(\Phi_{\mathrm{i}}(\vec{n}) \Rightarrow F(\vec{n}) = F_{\mathrm{i}}(\vec{n})\right)$$

for every $\mathrm{i} = 1, ..., \mathrm{n}$.

Proof. According to Theorem 5.4 there exists a characteristic function C_{i} for $\Phi_{\mathrm{i}}(\vec{n})$ for $\mathrm{i} = 1, ..., \mathrm{n}$. We define

$$F(\vec{n}) = F_1(\vec{n}) \cdot C_1(\vec{n}) + ... + F_{\mathrm{n}}(\vec{n}) \cdot C_{\mathrm{n}}(\vec{n})$$

Then F clearly satisfies the assertion. □

6. Cummulative operators and bounded quantifiers

We introduce two basic operators.

Definition 6.1. Let F be a $(\mathrm{k}+1)$-ary primitive recursive function. We define a *cummulative sum*

$$(\vec{n}, m) \mapsto \sum_{i<m} F(\vec{n}, i)$$

by primitive recursion

$$\sum_{i<0} F(\vec{n}, i) = 0, \sum_{i<\mathbf{S}(m)} F(\vec{n}, i) = F(\vec{n}, m) + \sum_{i<m} F(\vec{n}, i)$$

Definition 6.2. Let F be a $(\mathrm{k}+1)$-ary primitive recursive function. We define a *cummulative product*

$$(\vec{n}, m) \mapsto \prod_{i<m} F(\vec{n}, i)$$

by primitive recursion

$$\prod_{i<0} F(\vec{n}, i) = 1, \prod_{i<\mathbf{S}(m)} F(\vec{n}, i) = F(\vec{n}, m) \cdot \prod_{i<m} F(\vec{n}, i)$$

Remark 6.3. Let F be a $(\mathrm{k}+1)$-ary primitive recursive function. Then we use the following notation

$$\sum_{k\leq i<m} F(\vec{n}, i) \equiv \sum_{i<m\dot{-}k} F(\vec{n}, i+k), \prod_{k\leq i<m} F(\vec{n}, i) = \prod_{i<m\dot{-}k} F(\vec{n}, i+k)$$

Next result establishes basic properties of cummulative operators.

Fact 6.4. *Let F, G be $(\mathrm{k}+1)$-ary primitive recursive functions.*

(1) $\mathrm{PRA} \vdash (k < m) \Rightarrow \sum_{i<m} F(\vec{n}, i) = \sum_{i<k} F(\vec{n}, i) + \sum_{k\leq i<m} F(\vec{n}, i)$

(2) $\mathrm{PRA} \vdash \sum_{i<m} F(\vec{n}, i) + \sum_{i<m} G(\vec{n}, i) = \sum_{i<m} (F(\vec{n}, i) + G(\vec{n}, i))$

(3) $\mathrm{PRA} \vdash (k < m) \Rightarrow \prod_{i<m} F(\vec{n}, i) = \prod_{i<k} F(\vec{n}, i) \cdot \prod_{k\leq i<m} F(\vec{n}, i)$

(4) $\mathrm{PRA} \vdash \prod_{i<m} F(\vec{n}, i) \cdot \prod_{i<m} G(\vec{n}, i) = \prod_{i<m} (F(\vec{n}, i) \cdot G(\vec{n}, i))$

Proof. Left for the reader as an exercise. ☐

Remark 6.5. We use the following notation

$$\forall_{i<m} \Phi(\vec{n}, i) \equiv \forall_i (m \leq i \vee \Phi(\vec{n}, i)), \exists_{i<m} \Phi(\vec{n}, i) \equiv \exists_i (i < m \wedge \Phi(\vec{n}, i))$$

for every formula $\Phi(\vec{n}, i)$.

Theorem 6.6. *Let $\Phi(\vec{n}, i)$ be a primitive recursive formula. Then formulas*

$$\forall_{i<m} \Phi(\vec{n}, i), \exists_{i<m} \Phi(\vec{n}, i)$$

are primitive recursive.

For the proof we need the following results.

Lemma 6.6.1. *Let F be a $(\mathrm{k}+1)$-ary primitive recursive function. Then*

$$\mathrm{PRA} \vdash \forall_{i<m} F(\vec{n}, i) > 0 \Leftrightarrow \left(k \leq m \Rightarrow \prod_{i<k} F(\vec{n}, i) > 0\right)$$

Proof of the lemma. Suppose that $F(\vec{n}, i) > 0$ for every $i < m$. We prove

$$\left(k \leq m \Rightarrow \prod_{i<k} F(\vec{n}, i) > 0\right)$$

by induction on k. The base case is straightforward. Suppose that $k < m$ and $\prod_{i<k} F(\vec{n}, i) > 0$. Then

$$\prod_{i<\mathbf{S}(k)} F(\vec{n}, i) = F(\vec{n}, k) \cdot \prod_{i<k} F(\vec{n}, i) \underbrace{\geq}_{\text{induction hypothesis}} F(\vec{n}, k) \cdot 1 = F(\vec{n}, k) > 0$$

and hence the assertion follows by induction.

Suppose that

$$\left(k \leq m \Rightarrow \prod_{i<k} F(\vec{n}, i) > 0\right)$$

holds. Pick $k < m$. Then

$$F(\vec{n}, k) \cdot \prod_{i<k} F(\vec{n}, i) = \prod_{i<\mathbf{S}(k)} F(\vec{n}, i) > 0$$

Thus $F(\vec{n}, k) > 0$. Therefore, $F(\vec{n}, i) > 0$ for every $i < m$. □

Lemma 6.6.2. *Let F be a* $(\mathrm{k}+1)$*-ary primitive recursive function. Then*

$$\mathrm{PRA} \vdash \forall_{i<m} F(\vec{n}, i) = 0 \Leftrightarrow \left(k \leq m \Rightarrow \sum_{i<k} F(\vec{n}, i) = 0\right)$$

Proof of the lemma. Suppose that $F(\vec{n}, i) = 0$ for every $i < m$. We prove

$$\left(k \leq m \Rightarrow \sum_{i<k} F(\vec{n}, i) = 0\right)$$

by induction on k. The base case is straightforward. Suppose that $k < m$ and $\sum_{i<k} F(\vec{n}, i) = 0$. Then

$$\sum_{i<\mathbf{S}(k)} F(\vec{n}, i) = F(\vec{n}, k) + \sum_{i<k} F(\vec{n}, i) \underbrace{=}_{\text{induction hypothesis}} F(\vec{n}, k) = 0$$

and hence the assertion follows by induction.

Suppose that

$$\left(k \leq m \Rightarrow \sum_{i<k} F(\vec{n}, i) = 0\right)$$

holds. Pick $k < m$. Then

$$F(\vec{n}, k) + \sum_{i<k} F(\vec{n}, i) = \sum_{i<\mathbf{S}(k)} F(\vec{n}, i) = 0$$

Thus $F(\vec{n}, k) = 0$. Therefore, $F(\vec{n}, i) = 0$ for every $i < m$. □

Proof of the theorem. Let C be a characteristic function of $\Phi(\vec{n}, i)$.

According to Lemma 6.6.1 we have $\mathrm{sign}\left(\prod_{i<m} C(\vec{n}, i)\right) = 1$ if and only if $C(\vec{n}, i) = 1$ for all $i < m$. Thus

$$(\vec{n}, m) \mapsto \mathrm{sign}\left(\prod_{i<m} C(\vec{n}, i)\right)$$

is a characteristic function of $\forall_{i<m} \Phi(\vec{n}, i)$.

According to Lemma 6.6.2 we have $\mathrm{sign}\Big(\sum_{i<m} C(\vec{n},i)\Big) = 1$ if and only if there exists $i < m$ such that $C(\vec{n},i) = 1$. Thus

$$(\vec{n},m) \mapsto \mathrm{sign}\Big(\sum_{i<m} C(\vec{n},i)\Big)$$

is a characteristic function of $\exists_{i<m}\, \Phi(\vec{n},i)$. □

7. Induction schemata and bounded minimum

Next meta-theorem establishes equivalence of schemata expressing versions of mathematical induction for primitive recursive formulas.

Theorem 7.1. *The following schemata for primitive recursive formulas hold in* PRA*.*

(i) Induction schema.

$$\Big(\Phi(\vec{n},0) \wedge \forall_m\, (\Phi(\vec{n},m) \Rightarrow \Phi(\vec{n},\mathbf{S}(m)))\Big) \Rightarrow \forall_m\, \Phi(\vec{n},m)$$

(ii) Complete induction schema.

$$\forall_m\, (\forall_{i<m}\, \Phi(\vec{n},i) \Rightarrow \Phi(\vec{n},m)) \Rightarrow \forall_m\, \Phi(\vec{n},m)$$

(iii) Minimum principle.

$$\exists_m\, \Phi(\vec{n},m) \Rightarrow \exists_k\, (\Phi(\vec{n},k) \wedge \forall_m\, (\Phi(\vec{n},m) \Rightarrow k \leq m))$$

(iv) Finite descent principle.

$$\Big(\exists_m\, \Phi(\vec{n},m) \wedge \forall_m\, ((\Phi(\vec{n},m) \wedge m \neq 0) \Rightarrow \exists_{i<m}\, \Phi(\vec{n},i))\Big) \Rightarrow \Phi(\vec{n},0)$$

Proof. Theorem 5.4 implies that **(i)** holds in PRA. It suffices to prove that all schemata in the statement are equivalent in PRA.

Suppose that **(i)** holds. Pick some primitive recursive formula $\Phi(\vec{n},i)$. Define a formula

$$\Psi(\vec{n},m) \equiv \forall_{i<m}\, \Phi(\vec{n},i)$$

Then $\Psi(\vec{n},m)$ is primitive recursive according to Theorem 6.6. Next assuming

$$\forall_m\, (\forall_{i<m}\, \Phi(\vec{n},i) \Rightarrow \Phi(\vec{n},m))$$

we derive

$$\Psi(\vec{n},0) \wedge \forall_m\, (\Psi(\vec{n},m) \to \Psi(\vec{n},\mathbf{S}(m))$$

in PRA. Since **(i)** holds, we infer that $\forall_m\, \Psi(\vec{n},m)$. Thus $\forall_m \forall_{i<m}\, \Phi(\vec{n},i)$. Combining it with

$$\forall_m\, (\forall_{i<m}\, \Phi(\vec{n},i) \Rightarrow \Phi(\vec{n},m))$$

and logical axiom

$$\forall_m\, (\forall_{i<m}\, \Phi(\vec{n},i) \Rightarrow \Phi(\vec{n},m)) \Rightarrow (\forall_m \forall_{i<m}\, \Phi(\vec{n},i) \Rightarrow \forall_m\, \Phi(\vec{n},m))$$

we infer that $\forall_m\, \Phi(\vec{n},m)$. This proves that schema **(i)** implies schema **(ii)** in PRA.

Fix a primitive recursive formula $\Phi(\vec{n},m)$ and suppose that $\exists_m\, \Phi(\vec{n},m)$. Then it is not the case $\forall_m\, (\neg\Phi(\vec{n},m))$. Note that $\neg\Phi(\vec{n},m)$ is primitive recursive by Corollary 5.5. If **(ii)** holds for $\neg\Phi(\vec{n},m)$, then

$$\forall_m\, (\forall_{i<m}\, (\neg\Phi(\vec{n},i)) \Rightarrow (\neg\Phi(\vec{n},m)))$$

is not the case. Let z be such that

$$(\neg\, (\neg\Phi(\vec{n},k))) \wedge \forall_{i<k}\, (\neg\Phi(\vec{n},i))$$

Then

$$\Phi(\vec{n},k) \wedge \forall_{i<k} \left(\neg\Phi(\vec{n},i)\right)$$

Since

$$\forall_{i<k} \left(\neg\Phi(\vec{n},i)\right) \equiv \forall_i \left(k \leq i \vee \left(\neg\Phi(\vec{n},i)\right)\right)$$

and

$$\left(k \leq i \vee \left(\neg\Phi(\vec{n},i)\right)\right) \Leftrightarrow \left(\Phi(\vec{n},i) \Rightarrow k \leq i\right)$$

is a theorem of classical logic, we derive that

$$\Phi(\vec{n},k) \wedge \forall_m \left(\Phi(\vec{n},m) \Rightarrow k \leq m\right)$$

This completes the proof that schema **(ii)** implies schema **(iii)**.

Next suppose that **(iii)** holds. Let $\Phi(\vec{n},m)$ be a primitive recursive formula such that

$$\exists_m \Phi(\vec{n},m) \wedge \forall_m \left(\left(\Phi(\vec{n},m) \wedge m \neq 0\right) \Rightarrow \exists_{k<m} \Phi(\vec{n},k)\right)$$

Note that $\exists_m \Phi(\vec{n},m)$ and **(iii)** imply that there exists k such that

$$\Phi(\vec{n},k) \wedge \forall_m \left(\Phi(\vec{n},m) \Rightarrow k \leq m\right)$$

Assume that $k \neq 0$. Then $\forall_m \left(\left(\Phi(\vec{n},m) \wedge m \neq 0\right) \Rightarrow \exists_{i<m} \Phi(\vec{n},i)\right)$ implies that there exists $i < k$ such that $\Phi(\vec{n},i)$. Since $\forall_m \left(\Phi(\vec{n},m) \Rightarrow k \leq m\right)$, we derive that $k \leq i$. Together with PRA this implies that $k < k$ and this is a contradiction. Thus $k = 0$ and

$$\left(\exists_m \Phi(\vec{n},m) \wedge \forall_m \left(\left(\Phi(\vec{n},m) \wedge m \neq 0\right) \Rightarrow \exists_{i<m} \Phi(\vec{n},i)\right)\right) \Rightarrow \Phi(\vec{n},0)$$

Hence schema **(iii)** implies schema **(iv)** in PRA.

Finally assume that schema **(iv)** holds. Suppose that $\Phi(\vec{n},m)$ is a primitive recursive formula such that

$$\Phi(\vec{n},0) \wedge \forall_m \left(\Phi(\vec{n},m) \Rightarrow \Phi(\vec{n},\mathbf{S}(m))\right)$$

Suppose that $\neg\Phi(\vec{n},m)$ for some $m \neq 0$. Then PRA proves that there exists l such that $m = \mathbf{S}(l)$. It follows that $\Phi(\vec{n},l) \Rightarrow \Phi(\vec{n},\mathbf{S}(l))$. Combining this with $\neg\Phi(\vec{n},m)$ we infer that $\neg\Phi(\vec{n},l)$. Again invoking PRA we obtain

$$\left(\left(\neg\Phi(\vec{n},m)\right) \wedge m \neq 0\right) \Rightarrow \exists_{i<m} \left(\neg\Phi(\vec{n},i)\right)$$

By variable generalization

$$\forall_m \Bigg(\left(\left(\neg\Phi(\vec{n},m)\right) \wedge m \neq 0\right) \Rightarrow \exists_{i<m} \left(\neg\Phi(\vec{n},i)\right) \Bigg)$$

Suppose now that $\exists_m \left(\neg\Phi(\vec{n},m)\right)$. Then

$$\exists_m \left(\neg\Phi(\vec{n},m)\right) \wedge \forall_m \Bigg(\left(\left(\neg\Phi(\vec{n},m)\right) \wedge m \neq 0\right) \Rightarrow \exists_{i<m} \left(\neg\Phi(\vec{n},i)\right) \Bigg)$$

Since the negation of $\Phi(\vec{n},m)$ is primitive recursive by Corollary 5.5 and **(iv)** holds, we derive that $\neg\Phi(\vec{n},0)$. This leads to contradiction with $\Phi(\vec{n},0)$. Thus it is not the case that $\exists_m \left(\neg\Phi(\vec{n},m)\right)$ and hence $\forall_m \Phi(\vec{n},m)$. We deduce that schema **(iv)** implies schema **(i)** in PRA. □

Theorem 7.2. *Let $\Phi(\vec{n},i)$ be a primitive recursive formula. Then there exists a primitive recursive function*

$$(\vec{n},m) \mapsto \mu_{i<m}\left[\Phi(\vec{n},i)\right]$$

such that the following assertions hold.

(1) $\mathrm{PRA} \vdash \exists_{i<m} \Phi(\vec{n},i) \Rightarrow \Bigg(\Phi\left(\vec{n},\mu_{i<m}\left[\Phi(\vec{n},i)\right]\right) \wedge \forall_k \left(\Phi(\vec{n},k) \Rightarrow \mu_{i<m}\left[\Phi(\vec{n},i)\right] \leq k\right)\Bigg)$

(2) $\mathrm{PRA} \vdash \Bigg(\neg\exists_{i<m} \Phi(\vec{n},i)\Bigg) \Rightarrow \mu_{i<m}\left[\Phi(\vec{n},i)\right] = m$

Proof. Note that formulas

$$\Psi(\vec{n},k) = \forall_{i<k}\left(\neg\Phi(\vec{n},i)\right) \wedge \Phi(\vec{n},k),\ \Theta(\vec{n},m) = \forall_{i<m}\left(\neg\Phi(\vec{n},i)\right)$$

are primitive recursive by Corollary 5.5 and Theorem 6.6. Let C_Ψ, C_Θ be characteristic functions of $\Psi(\vec{n},k)$ and $\Theta(\vec{n},m)$, respectively. Then we set

$$\mu_{i<m}\left[\Phi(\vec{n},i)\right](\vec{n},m) = m \cdot C_\Theta(\vec{n},m) + \sum_{k<m} k \cdot C_\Psi(\vec{n},k)$$

Clearly $\mu_{i<m}\left[\Phi(\vec{n},i)\right]$ is primitive recursive.

We work in PRA and prove **(1)** and **(2)**.

Suppose that there exists $i < m$ such that $\Phi(\vec{n},i)$. According to the minimum principle (Theorem 7.1) there exists the smallest j such that $\Phi(\vec{n},j)$. Clearly $j < i$. Note that $\Psi(\vec{n},k)$ does not hold for all $k < j$ and $j < k$. Moreover, $\Theta(\vec{n},m)$ does not hold. Hence

$$\mu_{i<m}\left[\Phi(\vec{n},i)\right](\vec{n},m) = m \cdot C_\Theta(\vec{n},m) + \sum_{i<m} i \cdot C_\Psi(\vec{n},i) = j \cdot C_\Psi(\vec{n},j) = j$$

This completes the proof of **(1)**.

Next assume that $\Phi(\vec{n},i)$ does not hold for all $i < m$. Then $\Psi(\vec{n},k)$ does not hold for all $k < m$ and $\Theta(\vec{n},m)$ holds. Hence

$$\mu_{i<m}\left[\Phi(\vec{n},i)\right](\vec{n},m) = m \cdot C_\Theta(\vec{n},m) + \sum_{k<m} k \cdot C_\Psi(\vec{n},k) = m \cdot C_\Theta(\vec{n},m) = m$$

This proves **(2)**. □

Definition 7.3. Let $\Phi(\vec{n},i)$ be a primitive recursive formula. Then $\mu_{i<m}\left[\Phi(\vec{n},i)\right]$ is the *bounded minimum* of $\Phi(\vec{n},i)$.

8. Elementary number theory in Primitive Recursive Arithmetic

In this section we develop substantial amount of elementary number theory in PRA.

Definition 8.1.

$$n|m \equiv \exists_k\, m = k \cdot n$$

If $n|m$, then n *divides* m or n is a *divisor* of m.

Fact 8.2 [PRA]. $n|m \Leftrightarrow \exists_{k\leq m}\, m = k \cdot n$

Proof. Left for the reader as an exercise. □

Definition 8.3.

$$\text{Prime}(n) \equiv 1 < n \wedge \forall_m\left(m|x \Rightarrow (m = 1 \vee m = n)\right)$$

If $\text{Prime}(n)$, then n is a *prime*.

Fact 8.4 [PRA]. $\text{Prime}(n) \Leftrightarrow 1 < n \wedge \forall_{m<n}\ (m|n \Rightarrow m = 1)$

Proof. Left for the reader as an exercise. □

Proposition 8.5 [PRA]. *Let n be greater than 1. Then there exists prime m such that $m|n$.*

Proof. Fix $1 < n$. By Fact 8.2 formula
$$1 < m \wedge m|n$$
is primitive recursive. Since $1 < n \wedge n|n$, we apply the minimum principle to deduce that there exists the smallest m such that $1 < m \wedge m|n$. It suffices to prove that m is prime. Suppose that k is a divisor of m. If $1 < k < m$, then $k|n$ and this contradicts the minimality of m. Thus either $k = 1$ or $k = m$. □

Before proving the next very important theorem we need to introduce new primitive recursive function.

Definition 8.6. $0! = 1$, $(n+1)! = (n+1) \cdot n!$

Theorem 8.7 [Euclid, PRA]. *There exists primitive recursive function $n \mapsto p_n$ such that the following assertions hold.*

(1) *p_n is prime for every n.*

(2) *$p_n < p_{n+1}$ for every n.*

(3) *$n < p_n$ for every n.*

(4) *For every prime m there exists n such that $p_n = m$.*

We first need to prove the following results.

Lemma 8.7.1 [PRA]. $1 \leq m \leq n \Rightarrow m|n!$

Proof of the lemma. We prove this by induction on n. The base case is clear. Suppose that the result holds for some n. Pick $m \leq n+1$. If $m \leq n$, then $m|n!$ by induction hypothesis and hence
$$m|(n+1) \cdot n! = (n+1)!$$
If $m = n+1$, then $(n+1)! = m \cdot n!$. □

Lemma 8.7.2 [PRA]. *There exists a primitive recursive function G such that $G(n)$ is the smallest prime number greater than n.*

Proof of the lemma. According to Proposition 8.5 there exists prime m such that $m|n!+1$. Clearly $m \leq n!+1$. If $m \leq n$, then $1 \leq m \leq n$ and by Lemma 8.7.1 we derive that $m|n!$. Hence
$$m|(n!+1) \dot{-} n! = 1$$
This is contradiction. Thus $n < m \leq n!+1$. Next we define
$$G(n) = \mu_{i<n!+2}[\mathrm{Prime}(i) \wedge n < i]$$
According to Theorem 7.2 it follows that G is primitive recursive and $G(n)$ is the smallest prime number greater than n. □

Proof of the theorem. We define
$$p_0 = 2,\ p_{n+1} = G(p_n)$$
where G is primitive recursive function from Lemma 8.7.2. Since $n \mapsto p_n$ is obtained by primitive recursion, we derive that it is primitive recursive function. Clearly **(1)**, **(2)** and **(3)** hold.

Now we prove **(4)**. Let m be a prime number. Since $m < p_m$, there exists the smallest $k \leq m$ such that $m < p_k$. Note that $k \neq 0$ and hence there exists n such that $k = n+1$. Then $p_n \leq m < p_{n+1}$. If $p_n < m$, then
$$p_{n+1} = G(p_n) \leq m < p_{n+1}$$
and hence contradiction. Thus $m = p_n$. This proves **(4)**. □

Next we prove Euclidean division theorem in PRA.

Definition 8.8. $\mathrm{div}(n, m) = \mu_{i<n}\big[x < (i+1) \cdot m\big]$

Definition 8.9. $\mathrm{rem}(n, m) = n \dot{-} \mathrm{div}(n, m) \cdot m$

Theorem 8.10 [Euclidean division theorem, PRA]. *If $m \neq 0$, then*
$$n = \mathrm{div}(n, m) \cdot m + \mathrm{rem}(n, m)$$
and $\mathrm{rem}(n, m) < m$*. Moreover,* $\mathrm{div}(n, m)$ *and* $\mathrm{rem}(n, m)$ *are unique with respect to these properties.*

Proof. For $m \neq 0$ we have $n < (n+1) \cdot m$ and hence there exists $i < n+1$ such that $n < (i+1) \cdot m$. According to Theorem 7.2 the smallest i such that $n < (i+1) \cdot m$ is $\mathrm{div}(n, m)$. It follows that
$$\mathrm{div}(n, m) \cdot m \leq n < (\mathrm{div}(n, m) + 1) \cdot m$$
Hence $n = \mathrm{div}(n, m) \cdot m + \mathrm{rem}(n, m)$ and $\mathrm{rem}(n, m) < m$.

Assume that
$$q_1 \cdot m + r_1 = q_2 \cdot m + r_2$$
for some $r_1, r_2 < m$. Without loss of generality we may assume that $r_1 \leq r_2$. Then there exists r such that $r_2 = r + r_1$. Hence $q_1 \cdot m = q_2 \cdot m + r$. It follows that $q_2 \leq q_1$ and there exists q such that $q_1 = q_2 + q$. Then $q \cdot m = r$. If $q \neq 0$, then
$$m \leq q \cdot m = r \leq r_2 < m$$
This is a contradiction. Thus $q = 0$ and consequently $q_1 = q_2$ and $r_1 = r_2$. □

Now we prove Bézout theorem and Chinese remainder theorem in PRA. Both these results are related to the notion of relatively prime numbers.

Definition 8.11.
$$n \perp m \equiv (n \neq 0 \wedge m \neq 0) \wedge \forall_k \big((k|n \wedge k|m) \Rightarrow k = 1 \big)$$
If $n \perp m$, then n, m are *relatively prime*.

Fact 8.12 [PRA]. $n \perp m \Leftrightarrow \Big((n \neq 0 \wedge m \neq 0) \wedge \forall_{k \leq n} \big((k|n \wedge k|m) \Rightarrow k = 1\big) \Big)$

Proof. Left for the reader as an exercise. □

Theorem 8.13 [Bézout theorem, PRA]. *Let n, m be relatively prime. Then there exist a, b such that*
$$a \cdot n = b \cdot m + 1$$

The proof of Bézout theorem relies on two lemmas.

Lemma 8.13.1 [PRA]. *Let n, m be nonzero and let z be fixed. The following assertions are equivalent.*

- **(i)** *There exist a, b such that $a \cdot n = b \cdot m + k$.*
- **(ii)** *There exist a, b such that $a \cdot n + k = b \cdot m$.*

Proof of the lemma. Left for the reader as an exercise. □

Lemma 8.13.2 [PRA]. *Let n, m be nonzero and let k be smaller then n, m. Then the following assertions are equivalent.*

- **(i)** *There exist a, b such that $a \cdot n = b \cdot m + k$.*

(ii) *There exist $a < m$ and $b < n$ such that $a \cdot n = b \cdot m + k$.*

Proof of the lemma. Fix nonzero n, m and k not greater than n, m. Suppose that $a \cdot n = b \cdot m + k$. By Euclidean division theorem

$$a = q_a \cdot m + r_a, \; b = q_b \cdot n + r_b$$

for some q_a, q_b and $r_a < m, r_b < n$. Thus

$$q_a \cdot (n \cdot m) + r_a \cdot n = q_b \cdot (n \cdot m) + r_b \cdot m + k$$

Note that both $r_a \cdot n$ and $r_b \cdot m + k$ are smaller than $n \cdot m$. According to the uniqueness assertion of Euclidean division theorem we derive that $q_a = q_b$ and $r_a \cdot n = r_b \cdot m + k$. This completes the proof according to $r_a < m, r_b < n$. □

Proof of the theorem. We fix relatively prime n, m. Note that by definition n, m are nonzero.

Euclidean division theorem implies that there exists $k > 0$ smaller than n, m such that either

$$a \cdot n = b \cdot m + k$$

or

$$a \cdot n + k = b \cdot m$$

for some a, b. By Lemma 8.13.1 it follows that there exists $k > 0$ smaller than n, m such that

$$a \cdot n = b \cdot m + k$$

for some a, b. Applying the minimum principle and Lemma 8.13.2 we infer that there exists the smallest $k > 0$ such that $a \cdot n = b \cdot m + k$ for some a, b.

By Euclidean division algorithm there are q, r such that $m = q \cdot k + r$ and $r < k$. Then

$$(q \cdot a) \cdot n + r = (q \cdot b) \cdot m + q \cdot k + r = (q \cdot b + 1) \cdot m$$

Lemma 8.13.1 implies that $a \cdot n = b \cdot m + r$ for some a, b. If $0 < r$, then we obtain a contradiction with the minimality of k. Hence r is zero and k divides m.

Similarly by Euclidean division algorithm there are q, r such that $n = q \cdot k + r$ and $r < k$. Moreover, by Lemma 8.13.1 it follows that $a \cdot n + k = b \cdot m$ for some a, b. Then

$$(q \cdot b) \cdot m + r = (q \cdot a) \cdot n + q \cdot k + r = (q \cdot a + 1) \cdot n$$

This shows that $a \cdot n = b \cdot m + r$ for some a, b. If $r > 0$, then we obtain a contradiction with the minimality of k. Hence r is zero and k divides n.

Thus k divides both n and m. Since n and m are relatively prime, we infer that $k = 1$ and hence $a \cdot n = b \cdot m + 1$ for some a, b. □

Corollary 8.14 [Euclid's lemma, PRA]. $(n \perp m \wedge n \perp k) \Rightarrow n \perp m \cdot k$

Proof. According to Bézout theorem 8.13 there exist a, b, c, d such that $a \cdot m = b \cdot n + 1$ and $c \cdot k = d \cdot n + 1$. Then

$$(a \cdot c) \cdot (m \cdot k) = (b \cdot d + b + d) \cdot n + 1$$

Hence any common divisor of $m \cdot k$ and n divides 1. Thus n and $m \cdot k$ are relatively prime. □

Theorem 8.15 [Chinese remainder theorem, PRA]. *Let F, G be primitive recursive functions. Suppose that the following assertions hold.*

(1) *$F(t)$ are pairwise relatively prime for all $i < m$*

(2) *$G(i) < F(i)$ for all $i < m$.*

Then there exists $n < \prod_{i<m} F(i)$ *such that*
$$\text{rem}\,(n, F(i)) = G(i)$$
for all $i < m$.

For the proof we need the following result

Lemma 8.15.1 [PRA]. *Let F be a primitive recursive function and let n be such that $n \perp F(i)$ for all $i < m$. Then*
$$n \perp \prod_{i<m} F(i)$$

Proof of the lemma. The proof goes by induction on m by means of Corollary 8.14. The details are left for the reader. □

Proof of the theorem. Note that the statement can be expressed by primitive recursive formula. We apply induction on m. The base case is straightforward. Next suppose that

(1) $F(i)$ are pairwise relatively prime for all $i < m+1$

(2) $G(i) < F(i)$ for all $i < m+1$.

Then by induction hypothesis there exists k such that $k < \prod_{i<m} F(i)$ and
$$\text{rem}\,(k, F(i)) = G(i)$$
for all $i < m$. We set
$$\tilde{k} = \prod_{i<m+1} F(i) + k$$
Then
$$\text{rem}\,(\tilde{k}, F(i)) = G(i)$$
for all $i < m$ and $\tilde{k} \geq F(m) > G(m)$. By Lemma 8.15.1 and Bézout theorem 8.13 there exists a, b such that
$$a \cdot F(m) = b \cdot \prod_{i<m} F(i) + 1$$
We set
$$\tilde{n} = a \cdot F(m) \cdot (\tilde{k} \dot{-} G(m)) + G(m)$$
Then
$$\text{rem}\,(\tilde{n}, F(i)) = \text{rem}\left(b \cdot (\tilde{k} \dot{-} G(m)) \cdot \prod_{i<m} F(i) + G(m) + (\tilde{k} \dot{-} G(m)), F(i)\right) =$$
$$= \text{rem}\,\big(G(m) + (\tilde{k} \dot{-} G(m)), F(i)\big) \underbrace{=}_{\text{Proposition 4.11}} \text{rem}(\tilde{k}, F(i)) = G(i)$$
for all $i < m$. Moreover, we have
$$\text{rem}\,(\tilde{n}, F(m)) = \text{rem}\,\big(a \cdot F(m) \cdot (\tilde{k} \dot{-} G(m)) + G(m), F(m)\big) = \text{rem}(G(m), F(m)) = G(m)$$
Note that $n = \text{rem}\,\big(\tilde{x}, \prod_{i<m+1} F(i)\big)$ also satisfies these assertions and
$$n < \prod_{i<m+1} F(i)$$
This completes the proof. □

Finally, we prove fundamental theorem of arithmetic in PRA.

Definition 8.16. $n^0 = 1$, $n^{m+1} = n \cdot n^m$

Fact 8.17 [PRA]. *The following assertions hold.*

(1) $(n > 1 \wedge m < k) \Rightarrow n^m < n^k$

(2) $n > 1 \Rightarrow \forall_m\, m < n^m$

(3) $n^m \cdot n^k = n^{m+k}$

Proof. Left for the reader as an exercise. □

Definition 8.18. $\exp(n, m) = \mu_{i<n}[p_m^{i+1} \nmid n]$

First we note the following basic result.

Fact 8.19 [PRA]. $n \neq 0 \Rightarrow (p_m^i | n \Leftrightarrow i \leq \exp(n, m))$

Proof. We have $n < p_m^n$ by Fact 8.17 and hence $p_m^n \nmid n$. Thus $\exp(n, m)$ is the smallest i such that $p_m^{i+1} \nmid n$. It follows that

$$p_m^{\exp(n,m)} | n,\ p_m^{\exp(n,m)+1} \nmid n$$

Next note that

(1) If $p_m^i | n$, then $p_m^j | n$ for every $j \leq i$.

(2) If $p_m^i \nmid n$, then $p_m^j \nmid n$ for every $i \leq j$.

The assertion now follows from the combination of these observations. □

Fact 8.20 [PRA]. $n \neq 0 \Rightarrow \forall_i \exp(n, i) < n$

Proof. According to Fact 8.19 we infer that $p_i^{\exp(n,i)} | n$. Since $n \neq 0$, we derive $p_i^{\exp}(n, i) \leq n$. Combine this with Fact 8.17 to deduce that

$$\exp(n, i) < p_i^{\exp(n,i)} \leq n$$

This completes the proof. □

Theorem 8.21 [Unique prime factorization, PRA]. *The following assertions hold.*

(1) *If F is a primitive recursive function, then*

$$\exp\left(\prod_{i<m} p_i^{F(i)}, j\right) = \begin{cases} F(j) & \text{if } j < m \\ 0 & \text{if } m \leq j \end{cases}$$

(2) *If $1 \leq n \leq m$, then*

$$n = \prod_{i<m} p_i^{\exp(n,i)}$$

For the proof we need a series of lemmas.

Lemma 8.21.1 [PRA]. $j \neq i \Rightarrow p_j \nmid p_i^n$

Proof. We prove this by induction on n. The base case is clear. By induction hypothesis $p_j \nmid p_i^n$ for some n. Since p_i is prime, we derive that $p_j \nmid p_i$. Hence $p_j \perp p_i^n$ and $p_j \perp p_i$. Corollary 8.14 implies that $p_j \perp p_i^{n+1}$ by Fact 8.17. Thus $p_j \nmid p_i^{n+1}$ and this completes the proof. □

Lemma 8.21.2 [PRA]. *Let G be a primitive recursive function such that $G(j) = 0$. Then $p_j \nmid \prod_{i<m} p_i^{G(i)}$.*

Proof of the lemma. We prove this by induction on m. The base case is clear. By induction hypothesis $p_j \nmid \prod_{i<m} p_i^{G(i)}$ for some m. Note that Lemma 8.21.1 shows that $p_j \nmid p_m^{G(m)}$. Hence $p_j \perp \prod_{i<m} p_i^{G(i)}$ and $p_j \perp p_m^{G(m)}$. According to Corollary 8.14

$$p_j \perp \prod_{i<m+1} p_i^{G(i)}$$

Hence $p_j \nmid \prod_{i<m+1} p_i^{G(i)}$. □

Proof of the theorem. We set

$$G(i) = \begin{cases} 0 & \text{if } i = j \\ F(i) & \text{otherwise} \end{cases}$$

By Corollary 5.6 we infer that G is primitive recursive. Easy induction on m shows that

$$\prod_{i<m} p_i^{F(i)} = \begin{cases} \prod_{i<m} p_i^{G(i)} & \text{if } m \leq j \\ p_j^{F(j)} \cdot \prod_{i<m} p_i^{G(i)} & \text{if } j < i \end{cases}$$

Hence

$$\exp\left(\prod_{i<m} p_i^{F(i)}, j\right) = \begin{cases} \exp\left(\prod_{i<m} p_i^{G(i)}, j\right) & \text{if } m \leq j \\ \exp\left(p_j^{F(j)} \cdot \prod_{i<m} p_i^{G(i)}, j\right) & \text{if } j < m \end{cases}$$

Lemma 8.21.2 implies that $p_j \nmid \prod_{i<m} p_i^{G(i)}$ for all m. Thus

$$\exp\left(\prod_{i<m} p_i^{F(i)}, j\right) = \begin{cases} 0 & \text{if } m \leq j \\ F(j) & \text{if } j < m \end{cases}$$

This completes the proof of **(1)**.

Next we prove **(2)**. We prove the assertion by induction on n. The base case is straightforward. Suppose now that $n > 1$. We have two cases.

- n is prime. Then $n = p_j$ for some $j < n$ according to Euclid's Theorem 8.7 and

  $$\exp(n, i) = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$$

  Easy induction shows that

  $$j < m \Rightarrow n = \prod_{i<m} p_i^{\exp(n,i)}$$

  Since $j < n$, we infer that

  $$n \leq m \Rightarrow n = \prod_{i<m} p_i^{\exp(n,i)}$$

- There are $1 < k, t < n$ such that $n = k \cdot l$. Pick m no smaller than n. By induction hypothesis

  $$k = \prod_{i<m} p_i^{\exp(k,i)}, \; l = \prod_{i<m} p_i^{\exp(l,i)}$$

  According to Fact 6.4 we infer

  $$n = k \cdot l = \prod_{i<m} p_i^{\exp(k,i)+\exp(l,i)}$$

  Next **(1)** implies that $\exp(n, i) = \exp(k, i) + \exp(l, i)$ for every $i < m$. Thus

  $$n = \prod_{i<m} p_i^{\exp(n,i)}$$

This proves **(2)**. □

9. Formalization of sequences and course-of-values recursion

In this section we formalize finite sequences in PRA and prove important theorem on recursion scheme.

Definition 9.1.

$$\mathrm{Seq}(n) \equiv 1 \leq n \wedge \exists_{m<n} \forall_{i<n} \big((i < m \Rightarrow 1 \leq \exp(n,i)) \wedge (m \leq i \Rightarrow \exp(n,i) = 0) \big)$$

If $\mathrm{Seq}(n)$, then n is a *sequence*.

Definition 9.2.

$$\mathrm{lh}(n) = \begin{cases} \mu_{m<n} \left[\forall_{i<n} \left(m \leq i \Rightarrow \exp(n,i) = 0 \right) \right] & \text{if } \mathrm{Seq}(n) \\ 0 & \text{otherwise} \end{cases}$$

If n is a sequence, then $\mathrm{lh}(n)$ is the *length* of n.

Fact 9.3 [PRA]. *Let n be a sequence. Then the following assertions hold.*

(1) $\mathrm{lh}(n) < n$

(2) $1 < \exp(n,i)$ *for all* $i < \mathrm{lh}(n)$.

(3) $\exp(n,i) = 0$ *for all* $\mathrm{lh}(n) \leq i$.

Proof. We prove **(1)**. Note that $\mathrm{Seq}(n)$ implies $1 \leq n$. Theorem 8.21 implies

$$n = \prod_{i<n} p_i^{\exp(n,i)}$$

Uniqueness assertion of Theorem 8.21 implies that $\exp(n,i) = 0$ for all $n \leq i$. This proves that $\mathrm{lh}(n) < n$.

For the proof of **(2)**, **(3)** note that according to $\mathrm{Seq}(n)$ there exists m such that

$$(i < m \Rightarrow 1 \leq \exp(n,i)) \wedge (m \leq i \Rightarrow \exp(n,i) = 0)$$

for all $i < n$. Then $\mathrm{lh}(n) = m$ and the proof is completed. □

Definition 9.4.

$$n_i = \begin{cases} \exp(n,i) \dot{-} 1 & \text{if } \mathrm{Seq}(n) \\ 0 & \text{otherwise} \end{cases}$$

If n is a sequence and $0 \leq i < \mathrm{lh}(n)$, then n_i is the *i-th element* of n.

Corollary 9.5 [PRA]. $\mathrm{Seq}(n) \Rightarrow \forall_i \, n_i < n$

Proof. Follows immediately from Fact 8.20. □

Fact 9.6 [PRA]. $\mathrm{Seq}(n) \Rightarrow n = \prod_{i<\mathrm{lh}(n)} p_i^{1+n_i}$

Proof. From Fact 9.3 we deduce that

$$\exp(n,i) = \begin{cases} 1 + n_i & \text{if } i < \mathrm{lh}(n) \\ 0 & \text{if } \mathrm{lh}(n) \leq i \end{cases}$$

According to Theorem 8.21 we infer the assertion. □

Corollary 9.7 [PRA]. $(\mathrm{Seq}(n) \wedge \mathrm{Seq}(m)) \Rightarrow (n = m \Leftrightarrow \forall_i \, n_i = m_i)$

Proof. This is immediate consequence of Fact 9.6. □

Remark 9.8. Suppose that n is a sequence of length k. Then we write n by the juxtaposition notation

$$n_0 ... n_{k \dot{-} 1}$$

Next assume that m is another sequence of length l and $n_i = m$ for some $0 \leq i < k$. Then we write n by the following notation with parentheses

$$n_0 ... n_{i \dot{-} 1} \, (m_0 ... m_{l \dot{-} 1}) \, n_{i+1} ... n_{k \dot{-} 1}$$

Definition 9.9.

$$n \frown m = \begin{cases} \prod_{i < \mathrm{lh}(n)} p_i^{1+n_i} \cdot \prod_{\mathrm{lh}(n) \leq i < \mathrm{lh}(n) + \mathrm{lh}(m)} p_i^{1+m_i} & \text{if } \mathrm{Seq}(n) \text{ and } \mathrm{Seq}(m) \\ 0 & \text{otherwise} \end{cases}$$

If n and m are sequences, then $n \frown m$ is their *concatenation*.

Corollary 9.10 [PRA]. *Let n, m be sequences. Then $n \frown m$ is the unique sequence such that the following assertions hold.*

(1) $\mathrm{lh}(n \frown m) = \mathrm{lh}(n) + \mathrm{lh}(m)$

(2) $(n \frown m)_i = \begin{cases} n_i & \textit{if } i < \mathrm{lh}(n) \\ m_i & \textit{if } \mathrm{lh}(n) \leq i < \mathrm{lh}(n) + \mathrm{lh}(m) \end{cases}$

Proof. Follows from definition of $n \frown m$ and Fact 9.6. □

Corollary 9.11 [PRA]. *The following assertions hold.*

(1) $1 \frown n = n = n \frown 1$ *for every sequence n.*

(2) $(n \frown m) \frown k = x \frown (m \frown k)$

Proof. This is a consequence of Corollary 9.10. □

Definition 9.12.

$$n_{|i} = \begin{cases} n_0 ... n_{i \dot{-} 1} & \text{if } 1 < i < \mathrm{lh}(n) \\ 1 & \text{if } \mathrm{Seq}(n) \text{ and } i = 0 \\ 0 & \text{otherwise} \end{cases}$$

Definition 9.13. $n \triangleright m = \begin{cases} x \frown 2^{1+m} & \text{if } \mathrm{Seq}(n) \\ 0 & \text{otherwise} \end{cases}$

Definition 9.14. Let F be a $(\mathrm{k}+1)$-ary primitive recursive function. We define

$$\tilde{F}(\vec{n}, m) = F(\vec{n}, 0) ... F(\vec{n}, m \dot{-} 1)$$

for every m and every $\vec{n}$. Then $\tilde{F}$ is a *course-of-values* function for F.

Fact 9.15 [PRA]. *Let F be a $(\mathrm{k}+1)$-ary primitive recursive function. Then $\tilde{F}(\vec{n}, m)$ is a sequence of length m and*

$$\tilde{F}(\vec{n}, m)_i = F(\vec{n}, i)$$

for all $i < m$ and every $\vec{n}$.

Proof. This is a consequence of Theorem 8.21. □

Proposition 9.16 [course-of-values recursion, PRA]. *Let F be a* k*-ary primitive recursive and let G be a* $(\mathrm{k}+2)$*-ary primitive recursive function. Then there exists a* $(\mathrm{k}+1)$*-ary primitive recursive function H such that*

$$H(\vec{n},0) = F(\vec{n}),\ H(\vec{n},m+1) = G\left(\vec{n},m,\tilde{H}(\vec{n},m+1)\right)$$

for every $\vec{n}$ and m.

Proof. We define

$$\overline{G}(\vec{n},m,k) = k \triangleright G(\vec{n},m,k),\ \overline{F}(\vec{n}) = 2^{1+F(\vec{n})}$$

Then $\overline{G}$ is a $(\mathrm{k}+2)$-ary primitive recursive function and $\overline{F}$ is a k-ary primitive recursive function. Let $\overline{H}$ be a function obtained by primitive recursion from $\overline{G}$ and $\overline{F}$. Then

$$\overline{H}(\vec{n},0) = 2^{1+F(\vec{n})},\ \overline{H}(\vec{n},m+1) = \overline{H}(\vec{n},m) \triangleright G\left(\vec{n},m,\overline{H}(\vec{n},m)\right)$$

for every m and $\vec{n}$. Straightforward induction on m implies that

(1) $\overline{H}(\vec{n},m)$ is a sequence of length $m+1$ for every $\vec{n}$ and m.

(2) $\overline{H}(\vec{n},k)_i = \overline{H}(\vec{n},m)_i$ for all $i \leq k \leq m$.

We define $H(\vec{n},m) = \overline{H}(\vec{n},m)_m$. Then **(1)** and **(2)** imply that H is $(\mathrm{k}+1)$-ary primitive recursive function and $\tilde{H}(\vec{n},m+1) = \overline{H}(\vec{n},m)$. Thus

$$H(\vec{n},0) = F(\vec{n}),\ H(\vec{n},m+1) = G\left(\vec{n},m,\tilde{H}(\vec{n},m+1)\right)$$

for every $\vec{n}$ and m. $\square$

Next we study course-of-values recursion for primitive recursive formulas.

Definition 9.17. Let $\Psi(\vec{n},m,k)$ be a primitive recursive formula. A primitive recursive formula $\Phi(\vec{n},m)$ with characteristic function C_Φ such that

$$\mathrm{PRA} \vdash \Phi(\vec{n},m) \Leftrightarrow \Psi\left(\vec{n},m,\widetilde{C_\Phi}(\vec{n},m)\right)$$

is obtained from $\Psi(\vec{n},m,k)$ by *course-of-values recursion*.

Remark 9.18. Let $\Psi(\vec{n},m,k)$ be a primitive recursive formula. Then any two formulas obtained by course-of-values recursion from $\Psi(\vec{n},m,k)$ are provably equivalent in PRA.

Corollary 9.19. *Let $\Psi(\vec{n},m,k)$ be a primitive recursive formula. Then there exists a formula obtained by course-of-values recursion from $\Psi(\vec{n},m,k)$.*

Proof. Let C_Ψ be a characteristic function of $\Psi(\vec{n},m,k)$. Then Proposition 9.16 implies that there exists a primitive recursive function C such that

$$C(\vec{n},0) = C_\Psi(\vec{n},0,1)$$

and

$$C(\vec{n},m+1) = C_\Psi\left(\vec{n},m+1,\tilde{C}(\vec{n},m+1)\right)$$

Then $\Phi(\vec{n},m) \equiv C(\vec{n},m) = 1$ is a primitive recursive formula that satisfies the assertion. $\square$

10. Formalization of first-order logic in Primitive Recursive Arithmetic

In this section we formalize in PRA our presentation of first-order logic in [?]. This is arithmetization of first-order meta-theory.

Definition 10.1. We define *logical constants*.

$$\bot = 3, \to = 5, \land = 7, \lor = 9, \forall = 11, \exists = 13, = = 15$$

Definition 10.2. A *signature* consists of the following data.

- A primitive recursive formula $\mathrm{Rel}(r)$ and a primitive recursive function arr such that
$$\mathrm{PRA} \vdash \mathrm{Rel}(r) \Rightarrow (20 < r \land \mathrm{rem}(r, 6) = 1)\,,\ \mathrm{PRA} \vdash \mathrm{Rel}(r) \Rightarrow \mathrm{arr}(r) \neq 0$$
If $\mathrm{Rel}(r)$ holds, then r is a *relation symbol* and $\mathrm{arr}(r)$ is the *arity* of r.
- A primitive recursive formula $\mathrm{Fun}(f)$ and a primitive recursive function arf such that
$$\mathrm{PRA} \vdash \mathrm{Fun}(f) \Rightarrow (20 < f \land \mathrm{rem}(f, 6) = 3)$$
If $\mathrm{Fun}(f)$ holds, then f is a *function symbol* and $\mathrm{arf}(f)$ is the *arity* of f.

Definition 10.3.
$$\mathrm{Var}(x) \equiv (10 < x \land \mathrm{rem}(x, 6) = 5)$$
If $\mathrm{Var}(x)$, then x is a *variable*.

Fact 10.4 [PRA]. *At most one of the following assertions hold.*

(1) *n is a sequence.*

(2) *n is a logical constant.*

(3) *n is a relation symbol.*

(4) *n is a function symbol.*

(5) *n is a variable.*

Proof. Follows immediately from definitions and the fact that if n is a sequence, then either $n = 1$ or $2|n$. □

Definition 10.5. We define $\mathrm{Term}(t)$ by course-of-values recursion on the structure of t as follows.

(1) t is a variable.

(2) t is a sequence of the form $ft_1...t_n$ for $n + 1 = \mathrm{lh}(t)$, where f is a n-ary function symbol and $\mathrm{Term}(t_i)$ holds for all $1 \leq i \leq n$.

If $\mathrm{Term}(t)$, then t is a *term*.

Definition 10.6. $\mathrm{Eq}(\phi)$ holds if ϕ is a sequence of the form $t_1 = t_2$, where t_1 and t_2 are terms.

If $\mathrm{Eq}(\phi)$, then ϕ is an *equational formula*.

Definition 10.7. $\mathrm{RelAtomic}(\phi)$ holds if either ϕ is $\bot$ or it is a sequence of the form $rt_1...t_n$ for $n + 1 = \mathrm{lh}(\phi)$, where r is n-ary relation symbol and t_i is a term for every $1 \leq i \leq n$.

If $\mathrm{RelAtomic}(\phi)$, then ϕ is an *relational formula*.

Definition 10.8. $\mathrm{Atomic}(\phi)$ holds if either ϕ is relational formula or equational formula.

If $\mathrm{Atomic}(\phi)$, then ϕ is an *atomic formula*.

Remark 10.9. Let x be a variable. Then we write the sequences $\forall x$ and $\exists x$ as $\forall_x$ and $\exists_x$, respectively.

Definition 10.10. Let $\mathrm{Form}(\phi)$ be a primitive recursive formula obtained by course-of-values recursion on the structure on ϕ as follows.

(1) ϕ is an atomic formula.

(2) ϕ is a sequence of one of the forms

$$\phi_1 \wedge \phi_2,\ \phi_1 \vee \phi_2,\ \phi_1 \to \phi_2$$

and $\mathrm{Form}(\phi_1), \mathrm{Form}(\phi_2)$ hold.

(3) ϕ is a sequence of one of the forms

$$\forall_x\, \psi,\ \exists_x\, \psi$$

where x is a variable and $\mathrm{Form}(\psi)$ holds.

If $\mathrm{Form}(\phi)$, then ϕ is an *formula*.

Definition 10.11. Let $\mathrm{Var}(x, n)$ be a primitive recursive formula obtained by course-of-values recursion on the structure of n as follows.

(1) x is a variable and $n = x$.

(2) x is a variable and n is a term of the form $ft_1...t_k$ for $k + 1 = \mathrm{lh}(n)$ such that $\mathrm{Var}(x, t_i)$ for some $1 \le i \le k$.

(3) x is a variable and n is a relational formula of the form $rt_1...t_k$ for $k + 1 = \mathrm{lh}(n)$ such that $\mathrm{Var}(x, t_i)$ for some $1 \le i \le k$.

(4) x is a variable and n is an equational formula of the form $t_1 = t_2$ and either $\mathrm{Var}(x, t_1)$ or $\mathrm{Var}(x, t_2)$.

(5) x is a variable and n is a formula of one of the forms

$$\phi_1 \wedge \phi_2,\ \phi_1 \vee \phi_2,\ \phi_1 \to \phi_2$$

and either $\mathrm{Var}(x, \phi_1)$ or $\mathrm{Var}(x, \phi_2)$.

(6) x is a variable and n is a formula of one of the forms

$$\forall_y\, \psi,\ \exists_y\, \psi$$

and either $x = y$ or $\mathrm{Var}(x, \psi)$.

If $\mathrm{Var}(x, n)$, then x is a *variable* in n.

Definition 10.12. Let $\mathrm{Free}(x, \phi)$ be a primitive recursive formula obtained by course-of-values recursion on the structure of ϕ as follows.

(1) x is a variable and ϕ is an atomic formula and $\mathrm{Var}(x, \phi)$.

(2) x is a variable and ϕ is a formula of one of the forms

$$\phi_1 \wedge \phi_2,\ \phi_1 \vee \phi_2,\ \phi_1 \to \phi_2$$

and either $\mathrm{Free}(x, \phi_1)$ or $\mathrm{Free}(x, \phi_2)$.

(3) x is a variable and ϕ is a formula of one of the forms

$$\forall_y\, \psi,\ \exists_y\, \psi$$

and $x \neq y$ and $\mathrm{Free}(x, \psi)$.

If $\mathrm{Free}(x, \phi)$, then x is a *free* variable in ϕ.

Fact 10.13 [PRA]. *The following assertions hold.*

(1) *If* $\mathrm{Free}(x, \phi)$, *then* $\mathrm{Var}(x, \phi)$.

(2) *If* $\mathrm{Var}(x, n)$, *then* $x \le n$.

(3) *If* $\mathrm{Free}(x, \phi)$, *then* $x \le \phi$.

Proof. The proof goes by complete induction schema of Theorem 7.1. The details are left for the reader. □

Proposition 10.14 [PRA]. *There exists a primitive recursive function* free *such that* $\mathrm{free}(n)$ *is an increasing sequence of all free variables of* n.

Proof. Let G be a primitive recursive function given by

$$G(n, m, k) = \begin{cases} k \triangleright m & \text{if } \mathrm{Free}(m, n) \\ k & \text{otherwise} \end{cases}$$

Then

$$H(n, 0) = 1,\ H(n, m+1) = G(n, m, H(n, m))$$

Then by easy induction we prove that $H(n, m)$ is an increasing sequence consists of all free variables x of n such that $x < m$. Next define $\mathrm{free}(n) = H(n, n+1)$. Then $\mathrm{free}(n)$ is increasing sequence of all free variables of n. □

Definition 10.15. Let $\mathrm{Sub}(t, x, n)$ be a primitive recursive formula obtained by course-of-values recursion on the structure of n as follows.

(1) t is a term and x is a variable and n is term and $\mathrm{Var}(x, n)$.

(2) t is a term and x is a variable and n is an atomic formula and $\mathrm{Var}(x, n)$.

(3) t is a term and x is a variable and n is a formula of one of the forms

$$\phi_1 \wedge \phi_2,\ \phi_1 \vee \phi_2,\ \phi_1 \to \phi_2$$

and $\mathrm{Sub}(t, x, \phi_1)$ and $\mathrm{Sub}(t, x, \phi_2)$.

(4) t is a term and x is a variable and n is a formula of one of the forms

$$\forall_y\, \psi,\ \exists_y\, \psi$$

and $x \neq y$ and $\mathrm{Sub}(t, x, \psi)$.

(5) t is a term and x is a variable and n is a formula of one of the forms

$$\forall_x\, \psi,\ \exists_x\, \psi$$

If $\mathrm{Sub}(t, x, n)$, then t is *substitutable* for x in n.

Definition 10.16. Let sub be a primitive recursive function obtained by course-of-values recursion on the structure of n as follows.

$$\mathrm{sub}(t,x,n) = \begin{cases} x & \text{if } \mathrm{Term}(t) \wedge \mathrm{Var}(x) \wedge \mathrm{Var}(n) \wedge n \neq x \\ t & \text{if } \mathrm{Term}(t) \wedge \mathrm{Var}(x) \wedge n = x \\ n_0\mathrm{sub}(t,x,n_1)...\mathrm{sub}(t,x,n_{\mathrm{lh}(n)\dot{-}1}) & \text{if } \mathrm{Term}(n) \wedge \mathrm{lh}(n) > 0 \\ \mathrm{sub}(t,x,n_0) = \mathrm{sub}(t,x,n_2) & \text{if } \mathrm{Term}(t) \wedge \mathrm{Var}(x) \wedge \mathrm{Eq}(n) \\ n_0\mathrm{sub}(t,x,n_1)...\mathrm{sub}(t,x,n_{\mathrm{lh}(n)\dot{-}1}) & \text{if } \mathrm{Term}(t) \wedge \mathrm{Var}(x) \wedge \mathrm{RelAtomic}(n) \\ \mathrm{sub}(t,x,n_0) \wedge \mathrm{sub}(t,x,n_2) & \text{if } \mathrm{Sub}(t,x,n) \wedge n_1 = \wedge \\ \mathrm{sub}(t,x,n_0) \vee \mathrm{sub}(t,x,n_2) & \text{if } \mathrm{Sub}(t,x,n) \wedge n_1 = \vee \\ \mathrm{sub}(t,x,n_0) \to \mathrm{sub}(t,x,n_2) & \text{if } \mathrm{Sub}(t,x,n) \wedge n_1 = \to \\ \forall_{n_1} \mathrm{sub}(t,x,n_2) & \text{if } \mathrm{Sub}(t,x,n) \wedge n_0 = \forall \wedge n_1 \neq x \\ \exists_{n_1} \mathrm{sub}(t,x,n_2) & \text{if } \mathrm{Sub}(t,x,n) \wedge n_0 = \exists \wedge n_1 \neq x \\ n & \text{if } \mathrm{Sub}(t,x,n) \wedge n_0 = \forall \wedge n_1 = x \\ n & \text{if } \mathrm{Sub}(t,x,n) \wedge n_0 = \exists \wedge n_1 = x \\ 0 & \text{otherwise} \end{cases}$$

Fact 10.17 [PRA]. *The following assertions hold.*

(1) *If n is a term and t is substitutable for x in n, then* $\mathrm{sub}(t,x,n)$ *is a term.*

(2) *If n is a formula and t is substitutable for x in n, then* $\mathrm{sub}(t,x,n)$ *is a formula.*

Proof. Follows from the definition and complete induction schema of Theorem 7.1. □

Definition 10.18.

$$\begin{aligned} &\mathrm{PAxiom}(\phi) \equiv \exists_{\phi_1,\phi_2,\phi_3<\phi}\Big(\mathrm{Form}(\phi_1) \wedge \mathrm{Form}(\phi_2) \wedge \mathrm{Form}(\phi_3) \wedge \Big(\\ &\phi = ((\phi_1 \to (\phi_2 \to \phi_3)) \to ((\phi_1 \to \phi_2) \to (\phi_1 \to \phi_3))) \\ &\vee \phi = \phi_1 \to (\phi_2 \to \phi_1) \\ &\vee \phi = (\phi_1 \wedge \phi_2) \to \phi_1 \vee \phi = (\phi_1 \wedge \phi_2) \to \phi_2 \\ &\vee \phi = \phi_1 \to (\phi_2 \to (\phi_1 \wedge \phi_2)) \\ &\vee \phi = \phi_1 \to (\phi_1 \vee \phi_2) \vee \phi = \phi_2 \to (\phi_1 \vee \phi_2) \\ &\vee \phi = (\phi_1 \to \phi_3) \to ((\phi_2 \to \phi_3) \to ((\phi_1 \vee \phi_2) \to \phi_2)) \\ &\vee \phi = ((\phi_1 \to \perp) \to \perp) \to \phi_1 \Big)\Big) \end{aligned}$$

If $\mathrm{PAxiom}(\phi)$, then ϕ is a *propositional axiom*.

Definition 10.19.

$$\begin{aligned} &\mathrm{QAxiom}(\phi) \equiv \exists_{\phi_1,\phi_2,\phi_3,x,t<\phi}\Big(\mathrm{Form}(\phi_1) \wedge \mathrm{Form}(\phi_2) \wedge \mathrm{Var}(x) \wedge \mathrm{Term}(t) \wedge \Big(\\ &\phi = \forall_x (\phi_1 \to \phi_2) \to (\forall_x \phi_1 \to \forall_x \phi_2) \\ &\vee (\phi = \phi_1 \to \forall_x \phi_1 \wedge \mathrm{Free}(x,\phi_1)) \\ &\vee (\phi = \forall_x \phi_1 \to \mathrm{sub}(t,x,\phi_1) \wedge \mathrm{Sub}(t,x,\phi_1)) \\ &\vee (\phi = \mathrm{sub}(t,x,\phi_1) \to \exists_x \phi_1 \wedge \mathrm{Sub}(t,x,\phi_1)) \\ &\vee (\phi = \forall_x (\phi_1 \to \phi_2) \to (\exists_x \phi_1 \to \phi_2) \wedge (\neg\mathrm{Free}(x,\phi_2)))\Big)\Big) \end{aligned}$$

If $\mathrm{QAxiom}(\phi)$, then ϕ is a *quantifier axiom*.

Definition 10.20.

$$\mathrm{EqAxiom}(\phi) \equiv \exists_{t_1,t_2,t_3,x,t,\psi<\phi}\Big(\mathrm{Term}(t_1) \wedge \mathrm{Term}(t_2) \wedge \mathrm{Term}(t_3) \wedge$$
$$\mathrm{Var}(x) \wedge \mathrm{Term}(t) \wedge \mathrm{Atomic}(\psi) \wedge \Big($$
$$\phi = t_1 = t_1$$
$$\vee\, \phi = t_1 = t_2 \to t_2 = t_1$$
$$\vee\, \phi = t_1 = t_2 \to (t_2 = t_3 \to t_1 = t_3)$$
$$\vee\, \phi = t_1 = t_2 \to \mathrm{sub}(t_1, x, t) = \mathrm{sub}(t_2, x, t)$$
$$\vee\, \phi = t_1 = t_2 \to (\mathrm{sub}(t_1, x, \psi) \to \mathrm{sub}(t_2, x, \psi))\Big)\Big)$$

If $\mathrm{EqAxiom}(\phi)$, then ϕ is an *equational axiom*.

Definition 10.21. Let $\mathrm{Axiom}(\phi)$ be a primitive recursive formula obtained by course-of-values recursion on the structure of ϕ as follows.

(1) ϕ is a propositional axiom.

(2) ϕ is a quantifier axiom.

(3) ϕ is an equational axiom.

(4) ϕ is a formula of the form $\forall_x\, \psi$ and $\mathrm{Axiom}(\psi)$ holds.

If $\mathrm{Axiom}(\phi)$, then ϕ is a *logical axiom*.

Definition 10.22. $\mathrm{MP}\,(\phi, \psi, \rho) \equiv \mathrm{Form}(\phi) \wedge \mathrm{Form}(\psi) \wedge \rho = \phi \to \psi$

If $\mathrm{MP}\,(\phi, \psi, \rho)$, then ρ is obtained by *modus ponens* from ψ and ψ.

Definition 10.23. Let $\gamma(n)$ be a formula of PRA such that

$$\mathrm{PRA} \vdash \gamma(n) \Rightarrow \mathrm{Form}(n)$$

Then γ is a *predicate on formulas*.

Definition 10.24. Let $\gamma(n)$ be a predicate of formulas. Then we define

$$\mathrm{Proof}_\gamma(p) \equiv \mathrm{lh}(p) > 0 \wedge \forall_{i<\mathrm{lh}(p)}\big(\mathrm{Axiom}(p_i) \vee \gamma(p_i) \vee \exists_{j,k<i}\, \mathrm{MP}(p_j, p_k, p_i)\big)$$

If $\mathrm{Proof}_\gamma(p)$, then p is a *proof* from γ.

Definition 10.25. Let $\gamma(n)$ be a predicate of formulas. Then we define

$$\mathrm{Prf}_\gamma(p, \phi) \equiv \mathrm{Form}(\phi) \wedge \mathrm{Proof}_\gamma(p) \wedge p[\mathrm{lh}(p) \dot{-} 1] = \phi$$

If $\mathrm{Prf}_\gamma(p, \phi)$, then p is a *proof* of ϕ from γ.

Fact 10.26. *Let $\gamma(n)$ be a primitive recursive predicate of formulas. Then* $\mathrm{Proof}_\gamma(p)$ *and* $\mathrm{Prf}_\gamma(p, \phi)$ *are primitive recursive.*

Proof. Follows immediately from the definition. □

Definition 10.27. Let $\gamma(n)$ be a predicate of formulas. Then we define

$$\mathrm{Pr}_\gamma(\phi) \equiv \exists_p\, \mathrm{Prf}_\gamma(p, \phi)$$

The formula $\mathrm{Pr}_\gamma(\phi)$ is the *Gödel provability predicate* for γ.

11. Formalization of some metalogical results

We demonstrate now strength of PRA as a meta-theory by proving important meta-logical results.

Definition 11.1. Let γ be a primitive recursive predicate of formulas. Then

$$\begin{aligned}\operatorname{ant}(p,i) &= \mu_{j<i}\left[\operatorname{Proof}_\gamma(p) \wedge i < \operatorname{lh}(p) \wedge \exists_{k<i} \operatorname{MP}\left(p_j, p_k, p_i\right)\right] \\ \operatorname{cond}(p,i) &= \mu_{k<i}\left[\operatorname{Proof}_\gamma(p) \wedge \operatorname{ant}(p,i) < i \wedge \operatorname{MP}\left(p_{\operatorname{ant}(p,i)}, p_k, p_i\right)\right]\end{aligned}$$

Fact 11.2 [PRA]. *Let γ be a predicate on formulas and let p be a proof from γ. Fix $i < \operatorname{lh}(p)$. If neither γ holds for p_i nor p_i is an axiom, then* $\operatorname{ant}(p,i), \operatorname{cond}(p,i) < i$ *and*

$$p_{\operatorname{cond}(p,i)} = p_{\operatorname{ant}(p,i)} \to p_i$$

Proof. Left for the reader. □

Fact 11.3 [PRA]. *There exists a primitive recursive function F such that if ϕ is a formula, then $F(\phi)$ is a proof of $\phi \to \phi$.*

Proof. Fix a formula ϕ. Consider formulas

$$\begin{aligned}d_0(\phi) &= \phi \to (\phi \to \phi) \\ d_1(\phi) &= \phi \to ((\phi \to \phi) \to \phi) \\ d_2(\phi) &= (\phi \to ((\phi \to \phi) \to \phi)) \to ((\phi \to (\phi \to \phi)) \to (\phi \to \phi)) \\ d_3(\phi) &= (\phi \to (\phi \to \phi)) \to (\phi \to \phi) \\ d_4(n) &= \phi \to \phi\end{aligned}$$

Note that $d_0(\phi)d_1(\phi)d_2(\phi)d_3(\phi)d_4(\phi)$ is a proof. Indeed, $d_0(\phi), d_1(\phi), d_2(\phi)$ are propositional axioms, $d_3(\phi)$ is obtained by modus ponens from $d_2(\phi)$ and $d_1(\phi)$, $d_4(\phi)$ is obtained by modus ponens from $d_3(\phi)$ and $d_0(\phi)$. We define

$$F(\phi) = d_0(\phi)d_1(\phi)d_2(\phi)d_3(\phi)d_4(\phi)$$

By construction F satisfies the assertion. □

Let $\gamma(n)$ be a formulas predicate and let ϕ be a formula. We define

$$(\gamma \cup \phi)(n) \equiv \gamma(n) \vee n = \phi$$

We tacitly assume that writing symbol $\gamma \cup \phi$ entails that ϕ is a formula.

Theorem 11.4 [deduction theorem, PRA]. *Let $\gamma(n)$ be a primitive recursive predicate. Then the following assertions hold.*

- **(1)** *There exists a primitive recursive function F such that if p is a proof of ψ from $\gamma \cup \phi$, then $F(p,\phi)$ is a proof of $\phi \to \psi$.*
- **(2)** *There exists a primitive recursive function G such that if p is a proof of $\phi \to \psi$ from γ, then $G(p,\phi)$ is a proof of ψ from $\gamma \cup \phi$.*

For the proof we need some lemmas in which we construct some partial proofs.

Lemma 11.4.1 [PRA]. *There exists a primitive recursive function F such that if ψ is an axiom and ϕ is a formula, then $F(\psi,\phi)$ is a proof of $\phi \to \psi$.*

Proof of the lemma. Suppose that ψ is an axiom and ϕ is a formula. Consider formulas

$$\begin{aligned} &d_0(\psi) = \psi \\ &d_1(\psi, \phi) = \psi \to (\phi \to \psi) \\ &d_2(\psi, \phi) = \phi \to \psi \end{aligned}$$

Note that $d_0(\psi)d_1(\psi,\phi)d_2(\psi,\phi)$ is a proof. Indeed, $d_0(\psi)$ is an axiom, $d_1(\psi,\phi)$ is a propositional axiom and $d_2(\psi,\phi)$ is obtained by modus ponens from $d_1(\psi,\phi)$ and $d_0(\psi)$. We define

$$F(\psi,\phi) = d_0(\psi)d_1(\psi,\phi)d_2(\psi,\phi)$$

By construction F satisfies the assertion. □

Lemma 11.4.2 [PRA]. *Let $\gamma(n)$ be a primitive recursive predicate on formulas. There exists a primitive recursive function F such that if $\gamma(\psi)$ and ϕ is a formula, then $F(\psi,\phi)$ is a proof of $\phi \to \psi$ from γ.*

Proof of the lemma. Suppose that $\gamma(\psi)$ holds and ϕ is a formula. Consider formulas

$$\begin{aligned} &d_0(\psi) = \psi \\ &d_1(\psi, \phi) = \psi \to (\phi \to \psi) \\ &d_2(\psi, \phi) = \phi \to \psi \end{aligned}$$

Note that $d_0(\psi)d_1(\psi,\phi)d_2(\psi,\phi)$ is a proof from γ. Indeed, $\gamma\left(d_0(\psi)\right)$ holds, $d_1(\psi,\phi)$ is a propositional axiom and $d_2(\psi,\phi)$ is obtained by modus ponens from $d_1(\psi,\phi)$ and $d_0(\psi)$. We define

$$F(\psi,\phi) = d_0(\psi)d_1(\psi,\phi)d_2(\psi,\phi)$$

By construction F satisfies the assertion. □

Lemma 11.4.3 [PRA]. *There exists a primitive recursive function F such that if* $\mathrm{MP}(\psi_1,\psi_2,\psi_3)$ *and ϕ is a formula, then $F(\psi_1,\psi_2,\psi_3,\phi)$ is a proof of $\phi \to \psi_3$ from $\phi \to \psi_1, \phi \to \psi_2$.*

Proof of the lemma. Suppose that $\mathrm{MP}\left(\psi_1,\psi_2,\psi_3\right)$ and ϕ is a formula. Consider formulas

$$\begin{aligned} &d_0(\psi_1,\phi) = \phi \to \psi_1 \\ &d_1(\psi_2,\phi) = \phi \to \psi_2 \\ &d_2(\psi_1,\psi_2,\psi_3,\phi) = (\phi \to \psi_2) \to ((\phi \to \psi_1) \to (\phi \to \psi_3)) \\ &d_3(\psi_1,\psi_3,\phi) = (\phi \to \psi_1) \to (\phi \to \psi_3) \\ &d_4(\psi_3,\phi) = \phi \to \psi_3 \end{aligned}$$

Note that

$$d_0(\psi_1,\phi)d_1(\psi_2,\phi)d_2(\psi_1,\psi_2,\psi_3,\phi)d_3(\psi_1,\psi_3,\phi)d_4(\psi_3,\phi)$$

is a proof from $\phi \to \psi_1, \phi \to \psi_2$. Indeed, $d_2(\psi_1,\psi_2,\psi_3,\phi)$ is a propositional axiom by $\mathrm{MP}\left(\psi_1,\psi_2,\psi_3\right)$, $d_3(\psi_1,\psi_3,\phi)$ is obtained from $d_2(\psi_1,\psi_2,\psi_3,\phi)$ and $d_1(\psi_2,\phi)$ by modus ponens, and finally, $d_4(\psi_3,\phi)$ is obtained from $d_3(\psi_1,\psi_3,\phi)$ and $d_0(\psi_1,\phi)$ by modus ponens. We define

$$F(\psi_1,\psi_2,\psi_3,\phi) = d_0(\psi_1,\phi)d_1(\psi_2,\phi)d_2(\psi_1,\psi_2,\psi_3,\phi)d_3(\psi_1,\psi_3,\phi)d_4(\psi_3,\phi)$$

By construction F satisfies the assertion. □

Proof of the theorem. Let id be a primitive recursive function described by Fact 11.3. Let ax be a primitive recursive function descibed by Lemma 11.4.1. Let ga be a primitive recursive function described in Lemma 11.4.2. Let mp be a primitive recursive function described in Lemma 11.4.3. We construct F by course-of-values recursion on the structure of p. Suppose that p is a proof from $\gamma \cup \phi$, $n = \mathrm{lh}(p) \dot{-} 1$ and $p_n = \psi$. Then we set

$$F(p, \phi) = \begin{cases} \mathrm{id}(\phi, \phi) & \text{if } \mathrm{Prf}_{\gamma \cup \phi}(p, \psi) \wedge \phi = \psi \\ \mathrm{ax}(\psi, \phi) & \text{if } \mathrm{Prf}_{\gamma \cup \phi}(p, \psi) \wedge \mathrm{Axiom}(\psi) \\ \mathrm{ga}(\psi, \phi) & \text{if } \mathrm{Prf}_{\gamma \cup \phi}(p, \psi) \wedge \gamma(\psi) \end{cases}$$

and

$$F(p, \phi) = F(p_{|\mathrm{ant}(p,n)+1}, \phi) \frown F(p_{|\mathrm{cond}(p,n)+1}, \phi) \frown \mathrm{mp}(p_{\mathrm{ant}(p,n)}, p_{\mathrm{cond}(p,n)}, \psi, \phi)$$

if $\mathrm{ant}(p, n), \mathrm{cond}(p, m) < n$. Finally, if p is not a proof from $\gamma \cup \phi$, then we set $F(p, \phi) = 0$. According to Fact 11.3 and Lemmas 11.4.1, 11.4.2, 11.4.3 we infer that if p is a proof of ψ from $\gamma \cup \phi$, then $F(p, \phi)$ is a proof of $\phi \to \psi$ from γ. This proves **(1)**.

Next we define

$$G(p, \phi) = \begin{cases} p \frown \phi\psi & \text{if } \mathrm{Prf}_{\gamma}(p, \phi \to \psi) \\ 0 & \text{otherwise} \end{cases}$$

where ψ denotes $p[\mathrm{lh}(p) \dot{-} 1]$. Clearly if p is a proof of $\phi \to \psi$ from γ, then $G(p, \phi)$ is a proof of ψ from $\gamma \cup \phi$. This completes the proof of **(2)**. □

Theorem 11.5 [variable generalization, PRA]. *Let $\gamma(n)$ be a primitive recursive predicate on formulas. Then there exists a primitive recursive function F such that if p is a proof of some formula ϕ from γ and x is a variable that is not free in any formula for which predicate γ holds, then $F(p, x)$ is a proof of $\forall_x \phi$ from γ.*

For this meta-theorem we also need some constructions of proofs.

Lemma 11.5.1 [PRA]. *There exists a primitive recursive function F such that if ϕ is a formula and x is a variable that is not free in ϕ, then $F(\phi, x)$ is a proof of $\forall_x \phi$ from ϕ.*

Proof of the lemma. Suppose that ϕ is a formula and x is not free in ϕ. Consider formulas

$$d_0(\phi) = \phi$$
$$d_1(\phi, x) = \phi \to \forall_x \phi$$
$$d_2(\phi, x) = \forall_x \phi$$

Note that $d_0(\phi) d_1(\phi, x) d_2(\phi, x)$ is a proof from ϕ. Indeed, $d_0(\phi) = \phi$, $d_1(\phi, x)$ is a quantifier axiom, $d_2(\phi, x)$ is obtained by modus ponens from $d_1(\phi, x)$ and $d_0(\phi)$. We define

$$F(\phi, x) = d_0(\phi) d_1(\phi, x) d_2(\phi, x)$$

By construction F satisfies the assertion. □

Lemma 11.5.2 [PRA]. *There exists a primitive recursive function F such that if* $\mathrm{MP}(\psi_1, \psi_2, \psi_3)$ *and x is a variable, then $F(\psi_1, \psi_2, \psi_3, x)$ is a proof of $\forall_x \psi_3$ from $\forall_x \psi_1, \forall_x \psi_2$.*

Proof of the lemma. Suppose that MP (ψ_1, ψ_2, ψ_3) and x is a variable. Consider formulas

$$\begin{aligned}
&d_0(\psi_1, x) = \forall_x \psi_1 \\
&d_1(\psi_2, x) = \forall_x \psi_2 \\
&d_2(\psi_1, \psi_2, \psi_3, x) = \forall_x \psi_2 \to (\forall_x \psi_1 \to \forall_x \psi_3) \\
&d_3(\psi_1, \psi_3, x) = \forall_x \psi_1 \to \forall_x \psi_3 \\
&d_4(\psi_3, x) = \forall_x \psi_3
\end{aligned}$$

Note that

$$d_0(\psi_1, x)d_1(\psi_2, x)d_2(\psi_1, \psi_2, \psi_3, x)d_3(\psi_1, \psi_3, x)d_4(\psi_3, x)$$

is a proof from $\forall x\psi_1, \forall x\psi_2$. Indeed, $d_2(\psi_1, \psi_2, \psi_3, x)$ is a quantifier axiom by MP (ψ_1, ψ_2, ψ_3), $d_3(\psi_1, \psi_3, x)$ is obtained from $d_2(\psi_1, \psi_2, \psi_3, x)$ and $d_1(\psi_2, x)$ by modus ponens, and finally, $d_4(\psi_3, x)$ is obtained from $d_3(\psi_1, \psi_3, x)$ and $d_0(\psi_1, x)$ by modus ponens. We define

$$F(\psi_1, \psi_2, \psi_3, x) = d_0(\psi_1, x)d_1(\psi_2, x)d_2(\psi_1, \psi_2, \psi_3, x)d_3(\psi_1, \psi_3, x)d_4(\psi_3, x)$$

By construction F satisifes the assertion. □

Proof of the theorem. Consider formula

$$\mathrm{Free}_\gamma(x) \equiv \gamma(n) \Rightarrow \mathrm{Free}(x, n)$$

Since $\gamma(n)$ is primitive recursive, we derive that $\mathrm{Free}_\gamma(x)$ is a primitive recursive formula. Note that $\mathrm{Free}_\gamma(x)$ if x is not free in any formula for which predicate γ holds.

Let fr be a primitive recursive function described by Lemma 11.5.1. Let mp be a primitive recursive function descibed by Lemma 11.5.2. Suppose that x is not free in any formula for which γ holds. We construct F by course-of-values recursion on the structure on p. Let p be a proof from γ and let $n = \mathrm{lh}(p) \dot{-} 1$ and let $\phi = p_n$. We define

$$F(p, x) = \begin{cases} \forall_x \phi & \text{if } \mathrm{Axiom}(\phi) \\ \mathrm{fr}(\phi, x) & \text{if } \gamma(\phi) \end{cases}$$

and

$$F(p, x) = F(p_{|\mathrm{ant}(p,n)+1}, x) \frown F(p_{|\mathrm{cond}(p,n)+1}, x) \frown \mathrm{mp}(p_{\mathrm{ant}(p,n)}, p_{\mathrm{cond}(p,n)}, \phi, x)$$

if $\mathrm{ant}(p, n), \mathrm{cond}(p, n) < n$. Finally, we set $F(p, x) = 0$ for all other cases. According to Lemmas 11.5.1, 11.5.2 and the fact that axioms are closed under generalization we infer that if p is a proof of ϕ from γ and x is not free in γ, then $F(p, x)$ is a proof of $\forall_x \phi$ from γ. □

12. First-order arithmetic and Tarski's schema

In this section we introduce specific first-order language related to natural numbers. In particular, we are interested in formalizing in PRA Tarski's satisfaction relation for some fragment of first-order arithmetic.

Definition 12.1. The *language of arithmetic* is the first order language with identity whose signature consists of the following symbols:

$$\underbrace{o}_{\text{constant}}, \underbrace{<}_{\text{binary relation symbol}}, \underbrace{s}_{\text{unary function symbol}}, \underbrace{+, \cdot}_{\text{binary function symbols}}$$

We denote this language with **Ar**.

Next we formalize this language in PRA. For this we set

$$o = 21, s = 27, + = 33, \cdot = 39, <= 25$$

and

$$\mathrm{Rel}(r) \equiv r = 25,\ \mathrm{Func}(f) \equiv f = 21 \lor f = 27 \lor f = 33 \lor f = 39$$

We also set

$$\mathrm{arr}(25) = 2, \mathrm{arf}(21) = 0, \mathrm{arf}(27) = 1, \mathrm{arf}(33) = \mathrm{arf}(39) = 2$$

and zero otherwise. We can apply all results and constructions of previous two sections.

Remark 12.2. The important caveat regarding applications of the constructions of previous two sections to **Ar** is that we write its relation and function symbols in infix notation. This is natural convention for this symbols, since they are $<, +, \cdot$.

Remark 12.3. Let ϕ be a formula of the language of arithmetic and x, y be variables. Then we use notation

$$y \leq x \equiv y < x \vee y = x,\ \forall_{x<y}\, \phi \equiv \forall_x \left(y \leq x \vee \phi\right),\ \exists_{x<y}\, \phi \equiv \exists_x \left(x < y \wedge \phi\right)$$

Definition 12.4. $\overline{0} = o$, $\overline{n+1} = s\overline{n}$

The value of $\overline{n}$ is a *numeral*.

Fact 12.5 [PRA]. $\forall_n \operatorname{Term}(\overline{n}) \wedge \forall_n\, 0 < n < \overline{n}$

Proof. Easy induction on n. □

Definition 12.6. $\operatorname{Num}(n) \equiv \exists_m\, \overline{m} = n$

Fact 12.7 [PRA].

$$\operatorname{Num}(n) \Leftrightarrow \exists_{m<n}\, \overline{m} = n$$

Proof. Immediate consequence of Fact 12.5. □

Our goal is to define Tarski's semantics for some type of formulas of **Ar**. For this we need the notion of context of the evaluation or satisfaction.

Definition 12.8. $\operatorname{Env}(e) \equiv \operatorname{Seq}(e) \wedge \forall_{i<\operatorname{lh}(e)} \left(e_i = 0 \vee (\operatorname{Var}(i) \Rightarrow \operatorname{Num}(e_i))\right)$

If $\operatorname{Env}(e)$, then e is an *environment*.

Definition 12.9.

$$e[x/n] = \begin{cases} \left(e_x \triangleright \overline{n}\right) \frown \left(e_{x+1} \ldots e_{\operatorname{lh}(e) \dot{-} 1}\right) & \text{if } \operatorname{Env}(e) \wedge \operatorname{Var}(x) \wedge x < \operatorname{lh}(e) \\ 0 & \text{otherwise} \end{cases}$$

Definition 12.10.

$$\operatorname{Env}(e, n) \equiv \operatorname{Env}(e) \wedge n \leq \operatorname{lh}(e) \wedge (\operatorname{Term}(n) \vee \operatorname{Form}(n)) \wedge \forall_{x<\operatorname{lh}(e)} \left(\operatorname{Var}(x, n) \Rightarrow \operatorname{Num}(e_x)\right)$$

If $\operatorname{Env}(e, n)$, then e is an *environment* for n.

We define evaluation of terms in the environment.

Definition 12.11. Let eval be a primitive recursive function obtained by course-of-values recursion on the structure of t as follows.

$$\operatorname{eval}(e, t) = \begin{cases} e_t & \text{if } \operatorname{Env}(e, t) \wedge \operatorname{Var}(t) \\ \operatorname{eval}(e, t_1) + 1 & \text{if } \operatorname{Env}(e, t) \wedge \operatorname{Term}(t) \wedge t_0 = s \\ \operatorname{eval}(e, t_0) + \operatorname{eval}(e, t_2) & \text{if } \operatorname{Env}(e, t) \wedge \operatorname{Term}(t) \wedge t_1 = + \\ \operatorname{eval}(e, t_0) \cdot \operatorname{eval}(e, t_2) & \text{if } \operatorname{Env}(e, t) \wedge \operatorname{Term}(t) \wedge t_1 = \cdot \\ 0 & \text{otherwise} \end{cases}$$

Definition 12.12. Let $\mathrm{Rud}(\phi)$ be a primitive recursive formula obtained by course-of-values recursion on the structure of ϕ as follows.

(1) ϕ is an atomic formula.

(2) ϕ is a formula of one of the forms

$$\phi_1 \wedge \phi_2,\ \phi_1 \vee \phi_2,\ \phi_1 \to \phi_2$$

and $\mathrm{Rud}(\phi_1), \mathrm{Rud}(\phi_2)$ hold.

(3) ϕ is a formula of one of the forms

$$\forall_{x<y}\, \psi,\ \exists_{x<y}\, \psi$$

and $\mathrm{Rud}(\psi)$ holds.

If $\mathrm{Rud}(\phi)$ holds, then ϕ is a *rudimentary* formula.

Next we express Tarski's satisfaction schema for rudimentary formulas.

Definition 12.13. Let $\mathrm{Sat}(e, \phi)$ be a primitive recursive formula obtained by course-of-values recursion on the structure of ϕ as follows.

(1) $\mathrm{Env}(e, \phi)$ holds and ϕ is a formula of the form

$$t_1 = t_2$$

and t_1, t_2 are terms and $\mathrm{eval}(e, t_1) = \mathrm{eval}(e, t_2)$.

(2) $\mathrm{Env}(e, \phi)$ holds and ϕ is a formula of the form

$$t_1 < t_2$$

and t_1, t_2 are terms and $\mathrm{eval}(e, t_1) < \mathrm{eval}(e, t_2)$.

(3) $\mathrm{Env}(e, \phi)$ holds and ϕ is a formula of one of the forms

$$\phi_1 \wedge \phi_2,\ \phi_1 \vee \phi_2,\ \phi_1 \to \phi_2$$

and $\mathrm{Rud}(\phi_1), \mathrm{Rud}(\phi_2)$ hold and

$$\begin{aligned}&\mathrm{Sat}(e, \phi_1) \wedge \mathrm{Sat}(e, \phi_2),\\&\mathrm{Sat}(e, \phi_1) \vee \mathrm{Sat}(e, \phi_2),\\&\mathrm{Sat}(e, \phi_1) \Rightarrow \mathrm{Sat}(e, \phi_2),\end{aligned}$$

respectively.

(4) $\mathrm{Env}(e, \phi)$ holds and ϕ is a formula of one of the forms

$$\forall_{x<y}\, \psi,\ \exists_{x<y}\, \psi$$

and $\mathrm{Rud}(\psi)$ holds and

$$\begin{aligned}&\forall_{i<e_y} \mathrm{Sat}(e[y/i], \psi),\\&\exists_{i<e_y} \mathrm{Sat}(e[y/i], \psi),\end{aligned}$$

respectively.

If $\mathrm{Sat}(e, \phi)$, then a formula ϕ is *satisfied* in environment e.

We can also extend Tarski's schema to $\exists$-rudimentary formulas. The complication is that the formula expressing satisfaction is no longer primitive recursive.

Definition 12.14. $\mathrm{ExRud}(\phi)$ holds if ϕ is a formula of the form $\exists_x\, \psi$ and ψ is rudimentary.

If $\mathrm{ExRud}(\phi)$ holds, then ϕ is an $\exists$-*rudimentary* formula.

Definition 12.15. $\mathrm{ExSat}(e, \phi) \equiv \mathrm{Env}(e, \phi) \wedge \mathrm{ExRud}(\phi) \wedge \exists_n \mathrm{Sat}(e[\phi_1/n], \phi_2)$

References